



MS 491: MARKETING ANALYTICS

Indian Institute of Technology Gandhinagar

What Airbnb Pricing Patterns in Rio de Janeiro and Los Angeles County Mean for Managers

A Comparative Managerial Analytics Report

Group 13

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Executive Summary

Airbnb hosts and the platform itself face the same underlying question in every market: what makes one listing worth more than another, and does the answer travel from one city to the next? This report investigates that question using cleaned, currency-harmonized listings data from two structurally different markets: Rio de Janeiro, Brazil (35,376 listings after cleaning) and Los Angeles County, California (32,736 listings after cleaning). It moves from descriptive comparison through group-difference testing, correlation, and multivariate log-price regression, and finishes with a formal statistical test of where the two markets genuinely diverge.

A preliminary result frames everything that follows. Read in their native currencies, Rio's average nightly price of roughly R\$620 dwarfs Los Angeles County's \$177, and Rio would appear to be the premium market. Converted to a common currency, Rio averages about \$120 and Los Angeles County is in fact roughly 47% more expensive. Currency harmonisation is therefore not a footnote in this report but a first-order methodological decision, and every price figure quoted from Section 3 onward is already expressed in USD.

Four findings anchor the report. First, room type is the single most consistent price-associated characteristic in both markets: entire-home listings command a large premium over private rooms (147.1% in Rio and 145.7% in Los Angeles County), a relationship confirmed by highly significant analysis-of-variance tests in both cities and essentially unchanged after neighbourhood is statistically controlled for. Second, a formal cross-market interaction test shows that this entire-home premium is statistically indistinguishable between the two markets ($p \approx 0.353$), a genuine, tested similarity, not an assumption. Third, the hotel and shared-room categories behave differently: their price effects relative to private rooms do formally differ by market (hotel interaction $p \approx 0.013$; shared-room interaction $p < 0.001$), so a segmentation rule that works for entire-home listings does not automatically transfer to every room-type category. Fourth, review activity is negatively associated with price in both cities but far more steeply in Rio (-36.6% associated with each additional unit of reviews per month) than in Los Angeles County (-5.7%), and this difference is itself formally confirmed by a significant interaction term ($p < 0.001$).

Two descriptive layers sit alongside these tested results and are labelled as such throughout. Geography produces very wide price dispersion within each market: the highest-priced neighbourhoods with a usable sample size price at roughly 3.2× the city average in Rio (Joá) and roughly 4.0× the county average in Los Angeles County (Malibu, Beverly Crest), while the lowest-priced areas sit far below. And the two markets differ in what hosts actually supply: 71.6% of Rio's listings are entire homes against 63.2% in Los Angeles County, which matters because the room-type premium is the largest single effect in the analysis.

A fifth, exploratory but managerially important, thread concerns host portfolio scale. Hosts operating many listings are not a monolithic group: once room-type mix is held constant, Rio shows a real but modest, non-monotonic pattern, mid-size portfolios (2–9 listings) are associated with lower prices than single-listing hosts, while only the largest operators (10 or more listings) show a positive adjusted association (+10.6%). In Los Angeles County, the adjusted pattern is flat to mildly negative at every scale once room type is accounted for; the raw appearance of a "rise" at larger portfolio sizes in Los Angeles County is attributable to those hosts' room-type mix, not to scale itself. The one

host-scale pattern that does generalise across both markets is operational rather than financial: multi-listing hosts keep their properties bookable for substantially more days per year in both cities.

Throughout, every regression and ANOVA result is reported as a conditional association observed in a single cross-sectional snapshot of listings, not as a causal or predictive claim. Reviews per month and availability are treated strictly as engagement and calendar-activity proxies, never as measures of demand, bookings, or occupancy, and host-listings count is treated as an operational scale indicator, not a marker of professionalism. Roughly two-thirds of the variation in log-price remains unexplained by the variables available in this assignment's dataset ($R^2 \approx 0.28$ in Rio and ≈ 0.35 in Los Angeles County for the original models, rising modestly once neighbourhood is added), a reminder that unmeasured attributes such as size, amenities, and guest ratings likely carry real explanatory weight this analysis cannot capture.

Table ES1. Findings at a Glance

Question a manager asks	What the data shows	Evidence strength
Which market is the premium one?	Los Angeles County, about 47% higher mean price once converted to USD; the raw-currency figures point the opposite way.	Descriptive
What is the biggest price lever I can observe?	Room type. Entire homes price about 147% above private rooms in Rio and about 146% in Los Angeles County.	Confirmatory (ANOVA + regression)
Can I reuse one pricing rule in both markets?	For entire-home positioning, yes. For hotel and shared-room pricing and for review-based signals, no.	Confirmatory (pooled interaction test)
Does location override room type?	No. Controlling for neighbourhood shrinks the entire-home premium only modestly, to 136.0% and 140.2%.	Confirmatory (regression control)
Does guest engagement move with price?	Negatively in both markets, about six times more steeply in Rio on the regression coefficient.	Confirmatory (interaction $p < 0.001$)
Does running more listings mean charging more?	Only for Rio operators with 10+ listings (+10.6% adjusted). Flat to negative at every Los Angeles County tier.	Exploratory (per-market adjusted estimates)
How much of price does this explain?	About 28% of log-price variance in Rio and 35% in Los Angeles County; most variation is driven by attributes not in the dataset.	Model fit

A one-page orientation to the report. Each row is developed, with its full evidence, in the section indicated by the relevant table and figure references in the main body.

The report's core managerial contribution is Section 9, which separates what a platform-level regional manager and an individual host/listing manager should each take from this analysis. The central message for both: room-type positioning is a genuinely portable, cross-market benchmark, while review-based signals, hotel/shared-room pricing rules, and host-scale strategy are not; they must be calibrated separately for each market, not copied from one to the other.

1. Introduction & Marketing Problem

1.1 Business Context

Airbnb operates as a two-sided marketplace connecting independent hosts with travellers. Because hosts set their own prices, two listings in the same city, sometimes on the same street, can be priced very differently depending on the type of property, its location, how much guest engagement it has accumulated, and whether the host manages one listing or many. Price is one of the few outcomes in a marketplace that is observable for every listing, every day, and it is also one of the clearest expressions of how a market values a bundle of characteristics: location, space, service level, and reputation among them.

Treating price as the outcome variable to be explained, rather than a fixed input, turns a descriptive dataset into a managerial one: if we can show which observable listing characteristics are systematically associated with price, we can tell a host or a platform manager where to look first when they are trying to understand or benchmark a listing's position in its market.

1.2 Research Question and Objectives

This report addresses one specific version of that question: **which observable listing characteristics are most strongly associated with Airbnb price differences in Rio de Janeiro and Los Angeles County, and do these relationships look similar or different across the two markets, with "different" claimed only where a formal statistical interaction test supports it?**

The second half of that question matters as much as the first. It is easy to eyeball two coefficients from two separate regressions and conclude a market difference exists; it is a different and more defensible claim to test that difference directly and report only what survives the test. This report does the latter throughout, and Section 8 is devoted entirely to it.

Two markets were chosen deliberately rather than exhaustively: Rio de Janeiro, a coastal, leisure-oriented Brazilian market, and Los Angeles County, a large and economically diverse Californian market. They represent genuinely different price levels, currencies and, as the analysis will show, different internal pricing dynamics, which makes the cross-market comparison informative rather than a formality. The purpose of the comparison is not to determine which market is more expensive. It is to separate the price relationships that appear consistently in both places, and are therefore more likely to reflect a general pattern, from those that are specific to one market and would be misapplied if copied into the other.

Four observable listing characteristics are examined throughout:

- **Room type:** whether the listing is an entire home or apartment, a hotel room, a private room, or a shared room (Section 4).
- **Neighbourhood:** the listing's location within its city or county, examined both descriptively and as a statistical control (Section 5).
- **Review activity:** the number and frequency of guest reviews, used strictly as an observable indicator of guest engagement (Section 6).
- **Host scale:** whether a host manages a single listing or several, used strictly as an operational scale indicator (Section 7).

The analysis is written for two managerial readers. A platform or regional manager needs to know which pricing patterns are common enough across markets to standardize, and which require market-specific rules. An individual host or listing manager needs a benchmarking framework for

positioning a single property, not a formula that promises an "optimal" price. Neither reader can run a controlled experiment on their own portfolio; what this report offers instead is a disciplined read of the associations actually present in the data, with the honest caveats those associations carry.

2. Data & Methodology

2.1 Data Sources and Variables

The analysis uses Airbnb listings data for Rio de Janeiro and Los Angeles County, drawing on the variable set specified for this assignment: property and host identifiers, neighbourhood fields, latitude and longitude, room type, price, minimum nights, number of reviews, reviews per month, host listings count, and availability over 365 days. The "last review" date field was excluded from analysis, per the assignment's own instruction. Table 1 summarises the two source datasets.

Table 1. Overview of the Two Source Datasets

Dataset	Raw listings	Currency	Key variables used
Rio de Janeiro, Brazil	35,731	Brazilian Real (BRL, R\$)	price, room type, neighbourhood, reviews, availability, host listings count
Los Angeles County, California	33,078	US Dollar (USD, \$)	price, room type, neighbourhood, reviews, availability, host listings count

Both datasets share an identical variable structure, which is what makes a like-for-like cross-market comparison possible.

2.2 Sample and Cleaning

Both datasets were cleaned in two steps, applied separately within each market so that one market's price scale could not distort the other's. Listings with non-positive prices were removed first: 7 rows in Rio and 11 in Los Angeles County, all sitting at exactly zero, with no negative values in either file. The remaining listings were then trimmed at the inclusive 99th percentile of price to limit the influence of extreme outliers on descriptive statistics and regression estimates: this removed 348 Rio listings priced above roughly R\$8,852 and 331 Los Angeles County listings priced above roughly \$2,311. This left 35,376 usable Rio listings and 32,736 usable Los Angeles County listings, more than 99% of each original dataset retained.

Reviews per month is missing precisely when a listing has zero total reviews (confirmed as an exact correspondence in both files), so a listing with no reviews is treated as having a reviews-per-month rate of zero, the only mathematically coherent reading, not an arbitrary imputation choice.

A methodological note on the trimming rule. The assignment specifies removing the "top 1%" of prices without stating whether that percentile boundary itself should be included or excluded. For Rio, only the inclusive rule (keep price $\leq$ 99th percentile) reproduces the descriptive statistics used throughout this report; for Los Angeles County the inclusive and exclusive rules are numerically indistinguishable at the precision reported here. The inclusive rule is used consistently for both markets, and Section 10 reports a formal sensitivity check on how much the headline result moves under alternative trimming choices.

2.3 Currency Harmonisation

Rio prices are recorded in Brazilian reais; a conversion rate of 5.15 BRL/USD, reflecting rates observed in September 2026, is applied throughout so that every cross-market price comparison in this report is made in a common currency. This conversion is treated as a first-order methodological decision, not a footnote: comparing raw-currency figures directly would make Los Angeles County listings look far less expensive than a fair comparison supports and would, as Section 3 demonstrates, reverse the conclusion about which market is the premium one. Every price figure quoted in this report from Section 3 onward is already in USD; where a Rio figure appears in its original currency it is explicitly labelled BRL or R\$.

2.4 Analytical Approach

The report follows a descriptive-to-inferential progression, building evidence in stages rather than relying on any single number, chart, or model:

- **Descriptive statistics** (Section 3): mean, median, and dispersion of price in each market, establishing the overall price landscape before attempting to explain it.
- **Group comparison (ANOVA)** (Section 4): a formal test of whether average price differs significantly across room-type categories, run on both the raw and log price scales.
- **Correlation analysis** (Section 6): how price co-moves with review activity, minimum nights, availability, and host scale, taken one variable at a time.
- **Multivariate regression** (Sections 4 and 5): a log-price model considering all characteristics together, then a neighbourhood-augmented version testing whether the room-type story survives a location control.
- **Segment analyses** (Sections 5 to 7): descriptive neighbourhood benchmarking, review-status comparison, a supplementary nonlinear reviews specification, and a four-tier host-scale analysis.
- **Pooled cross-market interaction model** (Section 8): a single regression across both markets that formally tests which of the apparent cross-market differences are statistically real.

This progression matters because listing characteristics overlap in practice: a listing's room type, review history, and host's portfolio size are not independent of one another, so a conclusion drawn from one variable at a time can be actively misleading. Sections 6 and 7 each contain a concrete instance of exactly this problem, where a strong raw correlation shrinks dramatically or reverses once other factors are properly accounted for.

Table 2. Analytical Techniques Used, and What Each Is For

Technique	Question it answers	Where	Status
Descriptive statistics (mean, median, coefficient of variation)	What is the overall price level and spread in each market?	Section 3	Descriptive
One-way ANOVA (raw and log price)	Do average prices differ across room-type categories by more than sampling noise?	Section 4.2	Confirmatory
Multiple OLS regression on log price	Which characteristics remain associated with price once the others are held fixed?	Section 4.3	Confirmatory

Technique	Question it answers	Where	Status
Regression with neighbourhood indicators	Does the room-type result survive a location control?	Section 5.3	Confirmatory
Segment means and index comparison	Where do prices sit by neighbourhood, review status, and host tier?	Sections 5.1, 6.1, 7.1	Descriptive
Independent-samples t-tests	Are the reviewed/unreviewed and host-scale group gaps statistically reliable?	Sections 6.1, 7.1	Confirmatory
Pearson and Spearman correlation	How does price co-move with each variable on its own, and is that relationship driven by outliers?	Sections 6.2, 7.3	Exploratory
Log-transformed reviews specification	Is the review relationship better described as concave than linear?	Section 6.3	Exploratory
Room-type-adjusted tier analysis	Does host scale relate to price once room-type mix is removed?	Section 7.2	Confirmatory
Pooled regression with market interactions	Which cross-market differences are formally real rather than apparent?	Section 8	Confirmatory
Outlier-trimming sensitivity check	How much does the headline result depend on the cleaning rule?	Section 10.1	Exploratory / robustness

The status column is used consistently throughout the report. Descriptive analyses show where prices differ; confirmatory analyses test a stated hypothesis; exploratory analyses were developed after the core specification and carry a weaker standard of evidence.

Model form and standard errors. All regression models use natural-log price as the dependent variable, with coefficients converted to percentage effects using $100 \times (e^{\beta} - 1)$ rather than a raw $\beta \times 100$ approximation, the two diverge meaningfully for the larger coefficients in this dataset, and only the exponentiated form is reported here. Standard errors throughout the inferential sections use the heteroskedasticity-robust (HC1) form; a preliminary check found real, non-trivial heteroskedasticity in both markets' regression residuals, which the robust standard errors are designed to accommodate. Appendices A and B reproduce the same coefficients with classical (non-robust) standard errors as generated by the software, and are labelled accordingly; the main body's significance statements rest on the robust versions.

2.5 Software and Tools

All statistical work in this report (data cleaning, descriptive statistics, ANOVA, correlation, and every regression model) was carried out in Python, using pandas for data handling, scipy for hypothesis tests, and statsmodels for the ordinary least squares estimation. This plays the same role that Gretl played in the course's practical session: a statistical package applying standard formulas (OLS estimation, the F-test, the Pearson correlation coefficient) to the data. The underlying calculations are identical regardless of which package performs them. The complete, unedited software output for both original regression models is reproduced in Appendix D, so that every coefficient quoted in the main body can be traced back to its source.

2.6 How Every Result in This Report Should Be Read

Every regression coefficient, correlation, and test statistic that follows should be read as a conditional association, holding the other variables in the same model fixed, not as a causal effect. This is observational, cross-sectional data: no experiment, randomization, or repeated observation over time is available, and several of the "control" variables used throughout (reviews per month, availability, host-listings count) are plausibly determined jointly with price by the same host decisions, rather than independently of it. This caveat applies to every subsequent section and is not repeated in full each time, but it is never relaxed.

The report also distinguishes consistently between analyses that are descriptive and those that are confirmatory. The neighbourhood premium table in Section 5.1, the supply-mix comparison in Section 4.4, and the review-status comparison in Section 6.1 are descriptive: they show where prices differ, without isolating an independent effect. The regression and ANOVA results, and the pooled interaction test in Section 8, are inferential. Where a result is exploratory, developed after the core analysis, in response to a gap identified along the way; it is explicitly labelled as such in the table caption and in the surrounding text.

3. Market-Level Price Landscape

We begin with the simplest possible comparison: the average nightly price in each market. Rio's average price is approximately R\$620, while Los Angeles County's is approximately \$177. Read literally, these two numbers suggest Rio is more than three times as expensive. That comparison mixes two currencies and is not meaningful as it stands.

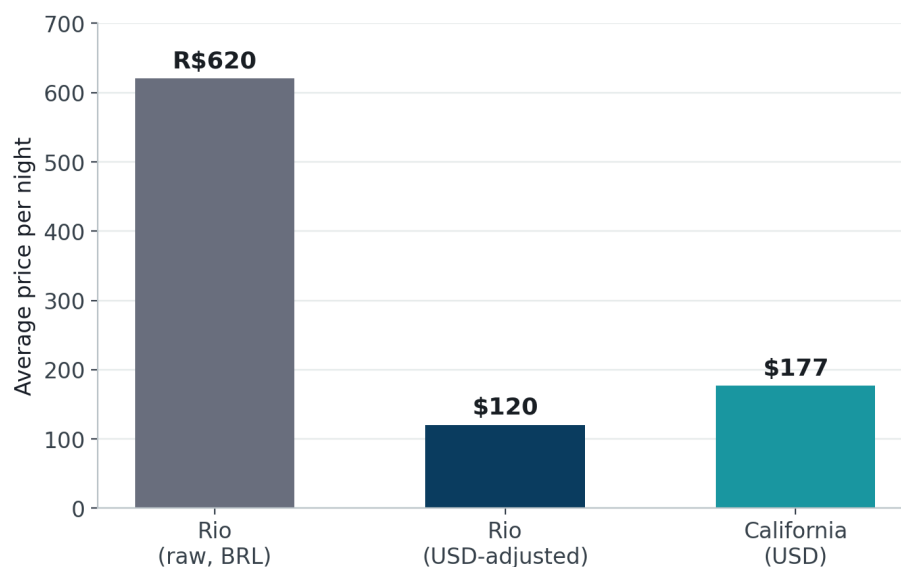


Figure 1. Average price per night: Rio's raw price in reais, Rio converted to US dollars, and Los Angeles County in US dollars.

Once Rio's average is converted at approximately 5.15 reais per dollar, it becomes about \$120 per night against \$177 in Los Angeles County. The comparison reverses completely: Los Angeles County is roughly 47% more expensive on average, not cheaper. This is the first substantive lesson of the report, and it is a methodological one; a cross-market price comparison is only as trustworthy as the currency conversion behind it.

The currency-adjustment catch

Judged on raw numbers alone, Rio would appear to be the premium market; judged correctly, it is Los Angeles County. Any manager or platform comparing pricing, targets, or performance across countries must convert to a common currency before drawing conclusions; otherwise the direction of the conclusion itself, not merely its magnitude, can be wrong.

Table 3. Market Descriptive Statistics

Market	N (cleaned)	Mean (USD)	Median (USD)	Coefficient of variation
Rio de Janeiro	35,376	\$120.45	\$58.25	1.66
Los Angeles County	32,736	\$177.01	\$107.00	1.33
Ratio (LA : Rio)	n/a	1.47×	1.84×	Rio is more dispersed

Cleaned sample after removing non-positive-price listings and trimming the top 1% of prices per market (inclusive 99th-percentile rule). Prices converted to USD at 5.15 BRL/USD where applicable.

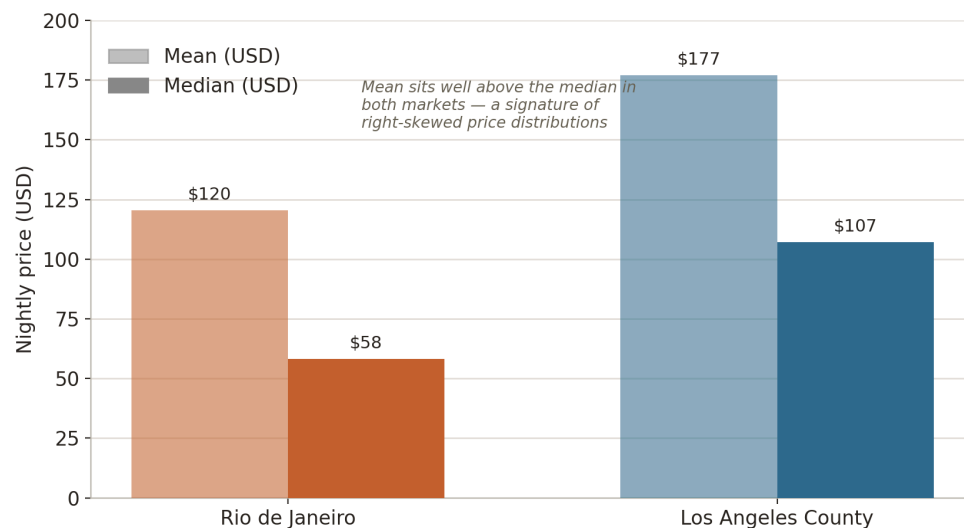


Figure 2. Mean vs. median listing price by market. Mean sits well above the median in both markets, a signature of right-skewed price distributions.

The mean is a misleading single number in both markets. Price is strongly right-skewed in both cities, a coefficient of variation of 1.66 in Rio and 1.33 in Los Angeles County, both well above the level at which a mean starts to be pulled upward by a comparatively small number of very high-priced listings. The median tells a more representative story of the "typical" listing: \$58.25 in Rio and \$107.00 in Los Angeles County. Notably, the gap is wider on the median (1.84×) than on the mean (1.47×), which says that the typical Los Angeles County listing is further above the typical Rio listing than the market averages alone suggest. The gap between mean and median in both cities, and its being larger in Rio, is itself informative: Rio's price distribution has a longer or heavier right tail relative to its typical listing than Los Angeles County's does.

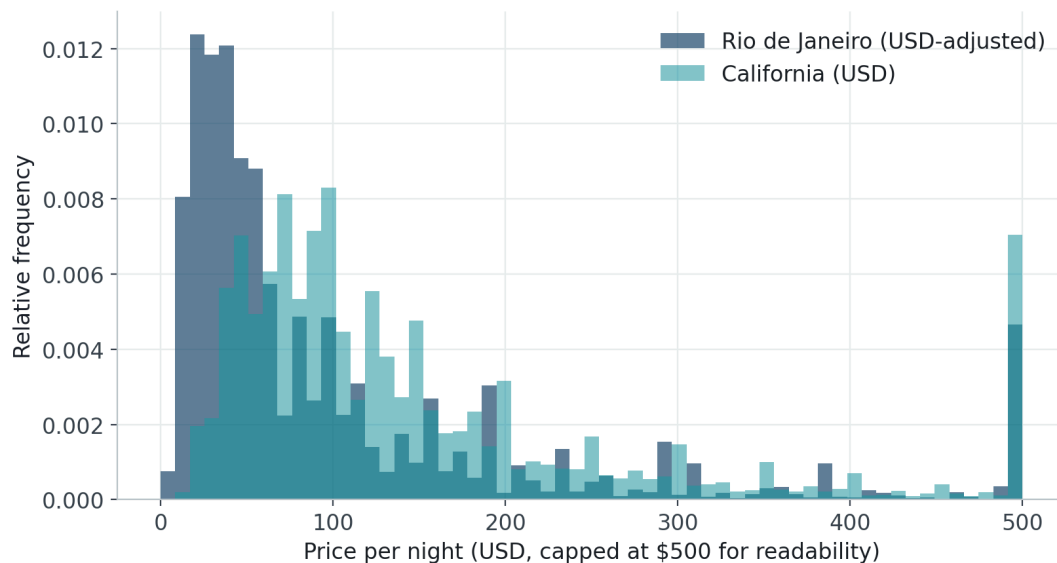


Figure 3. Distribution of nightly price, Rio de Janeiro (USD-adjusted) versus Los Angeles County (USD), capped at \$500 for readability.

The distributions make the same point visually. Rio's has a taller concentration of low-priced listings, while Los Angeles County's mid-range listings (roughly \$70 to \$200) make up a larger share of that market. In short, Los Angeles County has a higher typical price level, while Rio has a wider range of relative price outcomes around its own, lower centre.

What this shows and why it matters managerially. The headline "Los Angeles County is about 47% more expensive than Rio" is accurate only once both markets are converted to the same currency and only as a statement about central tendency, not about any individual listing. For a platform manager, this is a reminder that any cross-market price benchmark (a minimum price floor, a promotional threshold, a "typical listing" example used in marketing materials) should be built from medians, not means, and should never mix raw local-currency figures across markets. For a host, it is a caution against anchoring a price decision on an average pulled upward by a small number of atypical, high-priced listings in the same city.

This market-level landscape also sets up the rest of the report's approach. A coefficient of variation above 1.0 in both cities, meaning the standard deviation of price exceeds the mean itself, signals that a single market-wide "average price" is not a particularly useful benchmark for any specific listing; what a listing is, where it sits, and how it is used will matter far more than which city it happens to be in. That is exactly why Sections 4 through 7 each segment the market along a different observable dimension (room type, neighbourhood, review activity, and host scale in turn) rather than treating either market as a single undifferentiated price pool. The two markets' differing degree of skew (Rio's CV of 1.66 versus Los Angeles County's 1.33) is itself a first, modest hint that the two cities' internal price structures are not identical, a theme the report returns to with formal statistical backing in Section 8.

4. Room-Type Pricing

Room type divides listings into four categories (entire home or apartment, hotel room, private room, and shared room), and it turns out to be the most powerful and most consistent price signal available in this dataset, in both markets.

4.1 Descriptive Comparison

Table 4. Room-Type Price Comparison and Category Size

Room type	Rio: mean price (USD)	Rio: listings (n)	LA County: mean price (USD)	LA County: listings (n)
Entire home / apartment	\$147.18	25,324	\$231.19	20,678
Hotel room	\$117.05	164	\$149.05	146
Private room	\$52.41	9,119	\$86.66	10,750
Shared room	\$47.77	769	\$52.17	1,162

Mean nightly price (USD) and category size by room type, both markets. The hotel-room category is small in both markets (164 and 146 listings), which is worth bearing in mind when interpreting its coefficient later in this section.

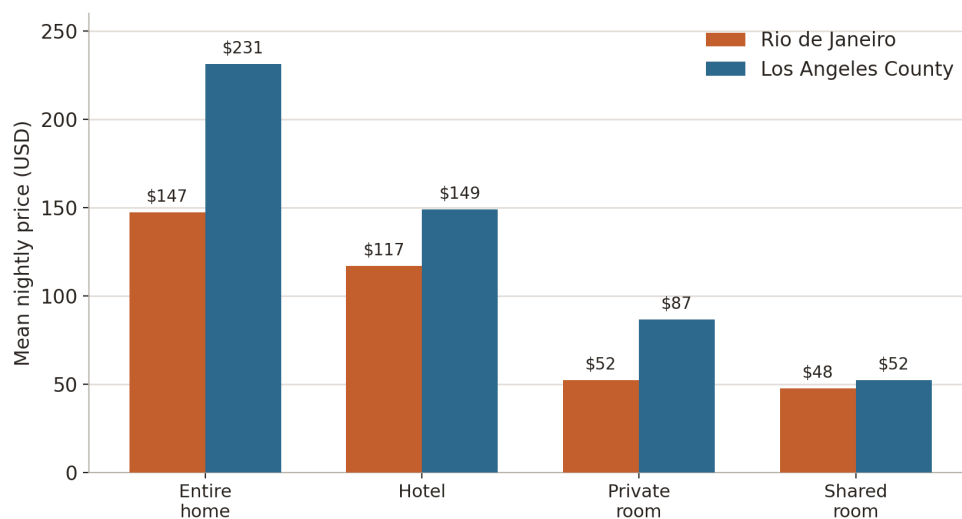


Figure 4. Room-type price comparison by market. Entire-home listings are the highest-priced category in both cities, by a wide margin.

Average price follows exactly the same order in both markets: entire home or apartment listings are the most expensive, followed by hotel rooms, then private rooms, then shared rooms. The ordering is intuitive, since an entire property offers more space and privacy than a shared room, but the managerial question is whether the gaps between categories are large enough, and reliable enough, to matter for pricing decisions.

4.2 Testing the Differences: Analysis of Variance

To test whether these differences are statistically real rather than a product of sampling variation, a one-way analysis of variance was run in each market, with room type as the grouping factor and price as the outcome. Formally, the hypotheses tested in each market are $H_0: \mu(\text{entire home}) = \mu(\text{hotel}) = \mu(\text{private room}) = \mu(\text{shared room})$, against H_1 : at least one room-type mean differs from the others. The test is run at the conventional 5% significance level.

Table 5. Room-Type ANOVA: Raw and Log-Price

Market	F (raw price)	F (log price)	df (between, within)	p-value
Rio de Janeiro	562.05	2,359	3, 35,372	< 0.001
Los Angeles County	1,111.97	5,180	3, 32,732	< 0.001

The log-price version is reported alongside the raw-price statistic because the classical equal-variance assumption is measurably violated on the raw price scale in both markets; both versions reject the null decisively. The full variance decomposition (sums of squares and mean squares behind each F-statistic) is reproduced in Appendix C.

Both F-statistics are large and both p-values fall far below the conventional 0.05 threshold, so the null hypothesis is rejected in both markets. Because the classical ANOVA assumes roughly equal price variance across room types, an assumption that does not fully hold here since price variance itself differs noticeably by room type. The same test was re-run on log-transformed price, which is less sensitive to that violation and gives an even larger, more decisive result in both markets (Rio $F \approx 2,359$; Los Angeles County $F \approx 5,180$). Both versions of the test agree: room type is not a marginal factor in either market's pricing, and the differences visible in Figure 4 are not sampling noise.

4.3 Room Type in the Regression Model

ANOVA establishes that room type matters on its own, but not whether the relationship survives once other listing characteristics are taken into account at the same time. The multivariate model estimated separately for each market takes the form:

$$\log(\text{Price}) = \beta_0 + \beta_1 \cdot \text{RoomType} + \beta_2 \cdot \text{ReviewsPerMonth} + \beta_3 \cdot \text{MinimumNights} + \beta_4 \cdot \text{Availability} + \beta_5 \cdot \text{HostListings} + \varepsilon$$

Table 6. Original Log-Price Regression, Both Markets

Variable (reference: private room)	Rio % effect	Significant?	LA County % effect	Significant?
Entire home / apartment	+147.1%	Yes ($p < 0.001$)	+145.7%	Yes ($p < 0.001$)
Hotel room	+31.3%	Yes ($p < 0.001$)	+68.1%	Yes ($p < 0.001$)
Shared room	-33.4%	Yes ($p < 0.001$)	-51.5%	Yes ($p < 0.001$)
Reviews per month (per unit)	-36.6%	Yes ($p < 0.001$)	-5.7%	Yes ($p < 0.001$)
Minimum nights (per night)	-0.005%	No ($p \approx 0.81$)	-0.16%	Yes ($p < 0.001$)
Availability (per day)	+0.004%	No ($p \approx 0.23$)	+0.058%	Yes ($p < 0.001$)
Host listings count (per listing)	+0.63%	Yes ($p < 0.001$)	-0.15%	Yes ($p < 0.001$)
R ² (log-price variance explained)	0.2825	n/a	0.3456	n/a
Observations	35,376	n/a	32,736	n/a

Percentage effects computed as $100 \times (e^{\beta} - 1)$, the exponentiated form, not a raw $\beta \times 100$ approximation. Overall model $F \approx 1,988.9$ (Rio) and $\approx 2,468.9$ (Los Angeles County), both $p < 0.001$. Full coefficient tables with standard errors and confidence intervals appear in Appendices A and B.

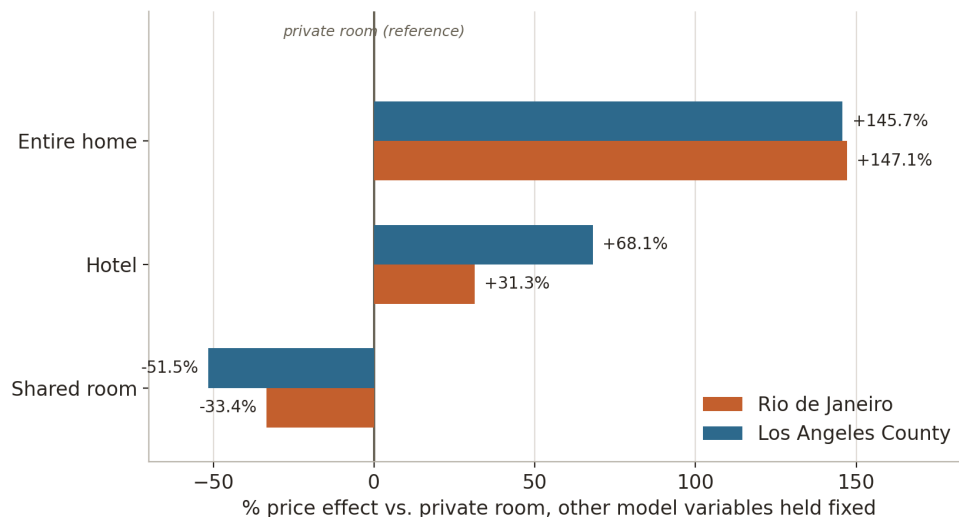


Figure 5. Room-type coefficients, original log-price regression. Percentage price effect vs. private room (reference), holding other model variables fixed.

The regression confirms that the room-type relationship survives holding other listing characteristics fixed, and that its size barely moves from the simple averages in Table 4. Using private room as the reference category, entire-home listings are associated with a 147.1% higher price in Rio and a 145.7% higher price in Los Angeles County. Hotel rooms show a smaller but still positive association in both markets (+31.3% in Rio, +68.1% in Los Angeles County), while shared rooms are associated with lower prices than private rooms in both cities (−33.4% in Rio, −51.5% in Los Angeles County). Together with reviews per month, minimum nights, availability, and host listings, these coefficients form a model that accounts for roughly 28% of the variance in log-price in Rio and roughly 35% in Los Angeles County, a meaningful share for a model built on a handful of structural characteristics, though far from a complete explanation (Section 10 returns to what that remaining variance likely reflects).

Two variables that are significant without being important. Minimum nights and availability are statistically significant in Los Angeles County and not in Rio, but in both markets their economic size is minimal: even a large change in either variable moves price by well under one percent. This is a useful illustration of the gap between statistical and practical significance on samples of this size, where very small effects clear conventional significance thresholds comfortably. They are retained in the model for completeness and as controls, but they are not a meaningful pricing lever in either market, and the report does not build any recommendation on them.

It is tempting, looking at 147.1% and 145.7%, to say the entire-home premium "looks the same" in both markets, and it is a reasonable-looking pattern. But this report does not treat that as established until it has been formally tested, which happens in Section 8; until then, this similarity should be read as suggestive, not confirmed.

4.4 Supply Mix: What Hosts Actually List

Price is only one side of the room-type story. The two markets also differ in what hosts choose to supply, which matters precisely because room type carries the largest price effect in the analysis.

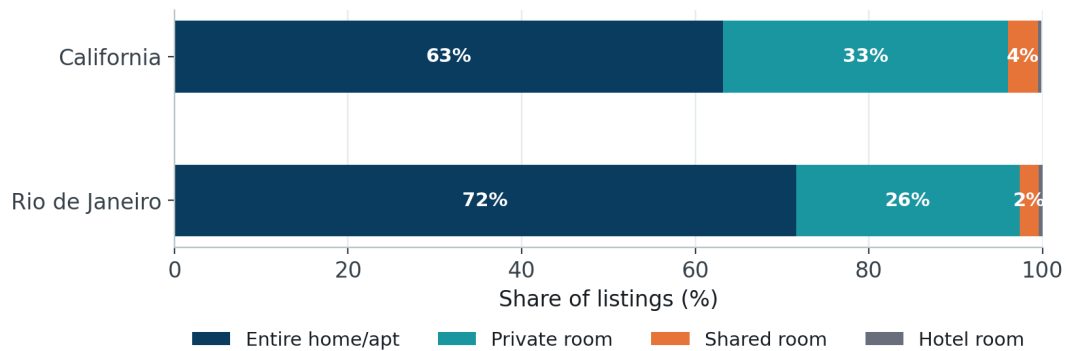


Figure 6. Composition of listings by room type, Rio de Janeiro versus Los Angeles County.

Entire-home listings make up 71.6% of Rio's supply, compared with 63.2% of Los Angeles County's; private rooms account for 25.8% of Rio's supply against 32.8% of Los Angeles County's. Rio's supply leans further toward whole-property, leisure-style stays, consistent with its profile as a coastal tourism destination, while Los Angeles County's market includes a somewhat larger share of private-room listings, consistent with a broader mix of long-term residents renting out spare rooms alongside dedicated short-term-rental properties. This is a compositional difference, not a price relationship: it does not affect the estimated entire-home premium, which is calculated within each market, but it does mean the two markets' overall price levels are being generated from differently weighted mixes of categories.

What this shows and why it matters managerially. Room type is the one segmentation variable in this analysis that behaves consistently and powerfully in both a Brazilian metro market and a US metro market with very different price levels, currencies, and (as later sections show) different secondary dynamics. For a platform manager building any kind of cross-market benchmarking tool, room type is the first and most reliable axis to segment on, and the supply-mix difference above is a reminder that identical segmentation logic will be applied to differently-shaped inventories in each market. For a host, it is the clearest reminder that a listing's basic category is doing a large share of the pricing work before anything else (location, reviews, or portfolio strategy) is considered; comparing a private room against a citywide average dominated by entire-home listings will systematically misstate what a competitive price looks like. The caveat that must travel with this finding: room type is an observed characteristic, not a lever a host can costlessly pull, and it is very likely correlated with unmeasured attributes this dataset does not capture (size, amenities, and build quality chief among them), so this is an association a host can use to benchmark, not a guarantee that reclassifying a listing would reproduce this price difference.

5. Geographic Variation

Room type is not the only source of price variation. Even among listings of the same category, location can place a property in a completely different price bracket. This section approaches geography in two distinct ways, and the distinction between them is important: Section 5.1 describes where prices are high and low, while Section 5.3 tests whether room type's apparent importance survives once location is statistically controlled for. The first is a benchmarking tool; the second is an inferential check. Neither substitutes for the other.

5.1 Descriptive Neighbourhood Benchmarking

Rio's listings span 156 distinct neighbourhoods and Los Angeles County's span 264. Table 7 shows the five highest-priced neighbourhoods in each market, restricted to neighbourhoods with at least 30 listings so that averages are not driven by a handful of properties.

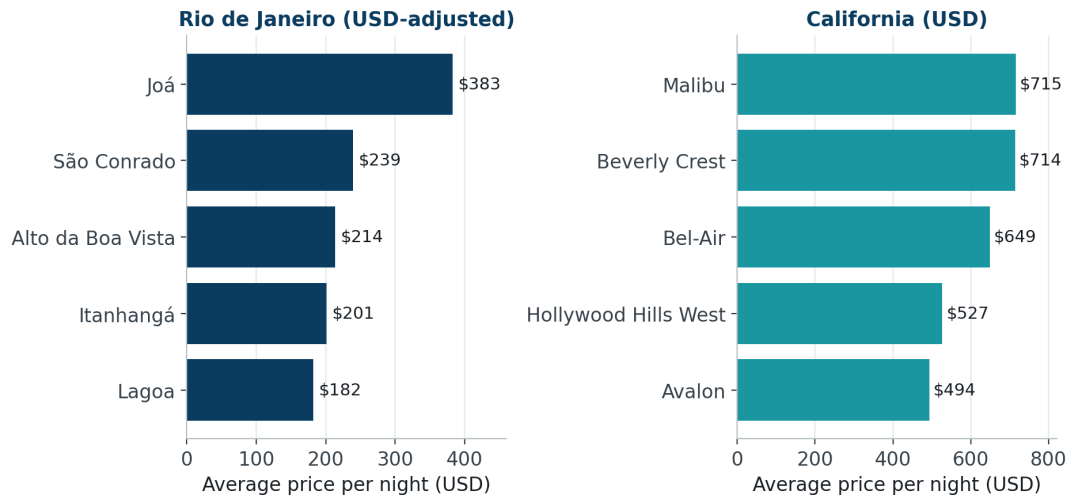


Figure 7. Five highest-priced neighbourhoods by average price per night, Rio de Janeiro (USD-adjusted) and Los Angeles County (USD).

Table 7. Highest-Priced Neighbourhoods and Their Premium Over Market Average

Rio neighbourhood	Avg. price (USD)	vs. city avg.	LA County neighbourhood	Avg. price (USD)	vs. county avg.
Lagoa	\$182	1.5×	Avalon	\$494	2.8×
Itanhangá	\$201	1.7×	Hollywood Hills West	\$527	3.0×
Alto da Boa Vista	\$214	1.8×	Bel-Air	\$649	3.7×
São Conrado	\$239	2.0×	Beverly Crest	\$714	4.0×
Joá	\$383	3.2×	Malibu	\$715	4.0×

Descriptive comparison. Neighbourhoods with at least 30 listings only. Premiums are relative to each market's own overall average (\$120 for Rio, \$177 for Los Angeles County).

At the other end of the scale, the lowest-priced neighbourhoods in Rio (Estácio and Méier among them, averaging roughly \$53 to \$54 after currency adjustment) and in Los Angeles County (Green Meadows and Lancaster, averaging \$26 to \$58) sit well below their market averages. The spread between the highest- and lowest-priced areas is wide in both cities, and materially wider in Los Angeles County: Joá prices at about 3.2 times Rio's overall average, while Malibu and Beverly Crest price at about 4.0 times Los Angeles County's.

A note on interpretation

This neighbourhood ranking is descriptive. It shows where prices are higher or lower; it does not isolate an independent "neighbourhood effect" separate from the type of properties that happen to be located there. A neighbourhood average combines whatever mix of room types, property

standards, and guest profiles exists in that area, and since room type carries the largest price effect in this analysis, an expensive neighbourhood may partly be an entire-home-heavy neighbourhood.

The regression in Section 5.3 addresses exactly this concern by including neighbourhood as a control alongside room type. What Table 7 supports on its own is that neighbourhood is a useful local benchmark, not that location independently causes the observed premium.

5.2 The Broader Neighbourhood Grouping

Both datasets provide neighbourhood at two levels of detail: a granular neighbourhood name, used throughout Section 5.1, and a broader neighbourhood grouping. In the Rio de Janeiro dataset the broader grouping field is empty for every listing and cannot be used, which is why the granular field is used for both markets to keep the comparison consistent. In the Los Angeles County dataset this field is fully populated and divides listings into three zones.

Table 8. Average Price by Broader Neighbourhood Zone, Los Angeles County

Zone (Los Angeles County only)	Avg. price (USD)	Listings (n)
City of Los Angeles	\$181	18,894
Other Cities	\$174	11,349
Unincorporated Areas	\$157	2,493

One-way ANOVA on zone: $F = 12.5$, $p < 0.001$. No equivalent breakdown exists for Rio de Janeiro because this field is not populated in the source data.

The zone-level differences within Los Angeles County are statistically significant but modest in size: listings inside the incorporated City of Los Angeles average about 15% more than listings in unincorporated areas, and noticeably smaller than either the room-type effect in Section 4 or the top-neighbourhood premiums in Table 7. This is a further instance of statistical significance without much practical weight: with more than 32,000 listings, a difference of this size clears the significance threshold easily while remaining too small to build a pricing rule around. The more useful geographic distinction in this market is the granular one.

5.3 Neighbourhood-Adjusted Regression

If room type's apparent importance were really a proxy for location, if entire-home listings simply happened to cluster in the more expensive parts of each city, then controlling for neighbourhood should sharply shrink or eliminate the entire-home effect. It does neither.

Table 9. Neighbourhood-Adjusted Regression

Market	Entire-home % (original)	Entire-home % (neighbourhood-adjusted)	R ² (original)	R ² (adjusted)
Rio de Janeiro	+147.1%	+136.0%	0.2825	0.3291
Los Angeles County	+145.7%	+140.2%	0.3456	0.3659

Adjusted model adds indicator variables for each market's 15 largest neighbourhoods plus a single pooled "other" category.

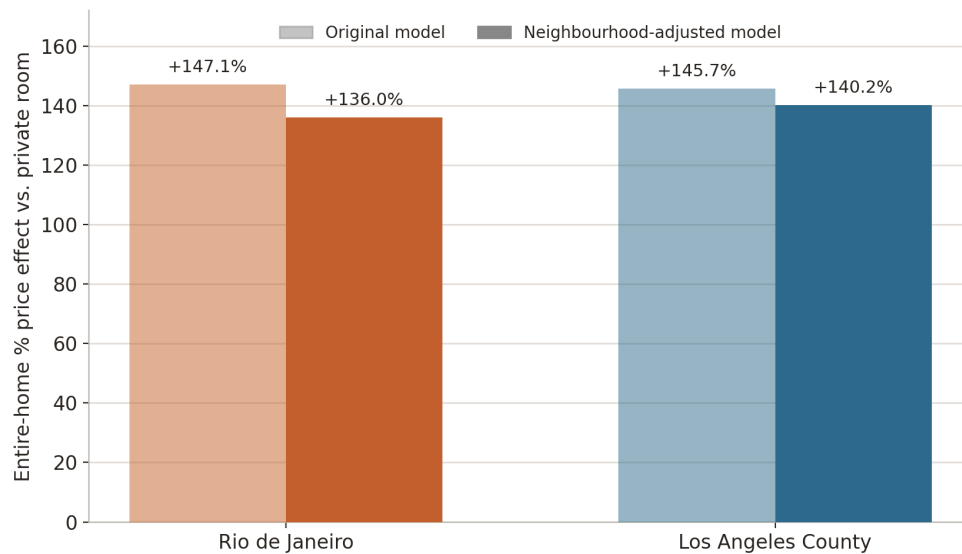


Figure 8. Entire-home premium: original vs. neighbourhood-adjusted. The premium shrinks modestly after controlling for neighbourhood; it does not disappear.

Adding neighbourhood indicators for each market's fifteen largest neighbourhoods, with every remaining neighbourhood pooled into a single "other" category, shrinks the entire-home premium only modestly: from 147.1% to 136.0% in Rio, and from 145.7% to 140.2% in Los Angeles County. Model fit improves by a moderate, not dramatic, amount; R^2 rises from 0.2825 to 0.3291 in Rio and from 0.3456 to 0.3659 in Los Angeles County. Part of the descriptive neighbourhood premium in Table 7 is therefore genuinely about location, and part of the room-type premium in Table 6 was indeed picking up geography, but only a modest part of each.

This top-15-neighbourhood specification is a deliberate, checked choice, not an arbitrary cutoff: the fifteenth-ranked neighbourhood in both markets still has several hundred listings behind it, comfortably above the point where an individual neighbourhood estimate becomes unstable, and the entire-home effect moves only modestly whether ten, fifteen, or twenty neighbourhoods are used individually. The "other" category itself should not be read as "the average neighbourhood"; it pools well over a hundred distinct, heterogeneous areas in each market, not a single typical location.

Individual named-neighbourhood premiums from this model are not the headline of this section, deliberately. Converting an individual neighbourhood's log-coefficient into a percentage requires the same exponential transform used for room type, $100 \times (e^{\beta} - 1)$. A raw coefficient read directly as a percentage will visibly understate the true premium for any sizable neighbourhood effect. The descriptive ranking in Table 7 is the more readable guide to which specific areas are expensive; the regression's contribution is the aggregate one: geography adds real explanatory power, but it does not explain away the room-type story.

What this shows and why it matters managerially. A platform manager tempted to build a location-only pricing benchmark should not: room type remains strongly associated with price even after neighbourhood is held fixed, so a benchmarking tool needs both dimensions, not one instead of the other. Equally, a listing should not be benchmarked against its entire city or county: a property in

Joá or Malibu operates in a materially different price environment than one in a typical neighbourhood. For a host, the practical sequencing implied here is to benchmark first against comparable room types, then refine within a neighbourhood tier, not the reverse. As with every regression result in this report, this is a statistical control, not proof that location itself causes the observed price differences; unmeasured neighbourhood characteristics such as proximity to specific attractions, safety perception, and transit access are not captured by a neighbourhood name alone.

6. Review Activity and Price

Reviews accumulate only after a guest has booked and stayed, so review activity is the closest proxy this dataset offers for how actively a listing is being engaged with by guests. It is not a measure of bookings, occupancy, or demand, and it is not treated as one anywhere in this report. With that proxy status firmly in mind, its relationship with price is one of the more striking asymmetries in the whole analysis. This section examines it in three complementary ways: a binary review-status comparison, the continuous reviews-per-month relationship in correlation and regression, and a supplementary nonlinear specification.

6.1 Review Status: Reviewed versus Unreviewed Listings

The simplest cut compares listings with at least one recorded review to listings with none, using a price index in which 100 represents each market's own average price, so that Rio and Los Angeles County can be compared on the same scale despite their different price levels.

Table 10. Price Index by Review Status

Market	No recorded reviews (index)	Has recorded reviews (index)	Gap (points)	Listings with no reviews	t-statistic
Rio de Janeiro	157.7	58.9	98.8	41.6%	-50.6 (p < 0.001)
Los Angeles County	125.3	93.3	32.0	20.9%	-14.2 (p < 0.001)

An index value above 100 means the group is priced above its own market's average. Independent-samples t-tests confirm the reviewed/unreviewed difference is statistically significant in both markets.

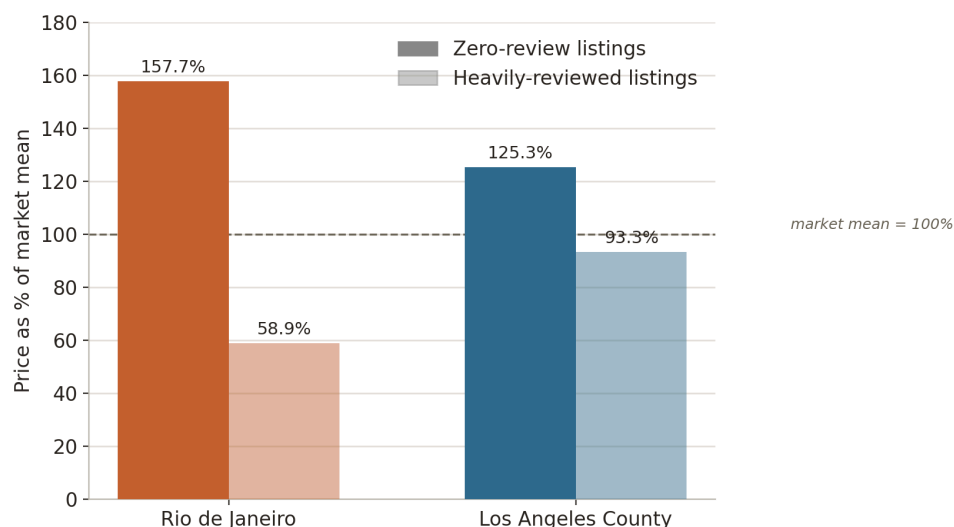


Figure 9. Review activity vs. price tier, relative to market mean. Zero-review listings sit well above the market mean in both cities; the gap to heavily-reviewed listings is much wider in Rio.

In both markets, listings with no recorded reviews are priced well above the market average while listings with recorded reviews are priced below it, and the gap is roughly three times larger in Rio (98.8 index points) than in Los Angeles County (32.0 points). The share of listings with no reviews at all also differs sharply between the two markets: 41.6% in Rio against 20.9% in Los Angeles County, which means the high-priced, unreviewed group is not a small fringe in Rio but a substantial fraction of its supply.

6.2 Reviews per Month: Correlation and Regression

Before combining every characteristic into a single model, it is worth seeing how price relates to each variable individually, using the Pearson correlation coefficient. This coefficient ranges from -1 to $+1$: positive values mean the variable tends to rise with price, negative values mean it tends to move opposite to price, and values near zero indicate little linear relationship.

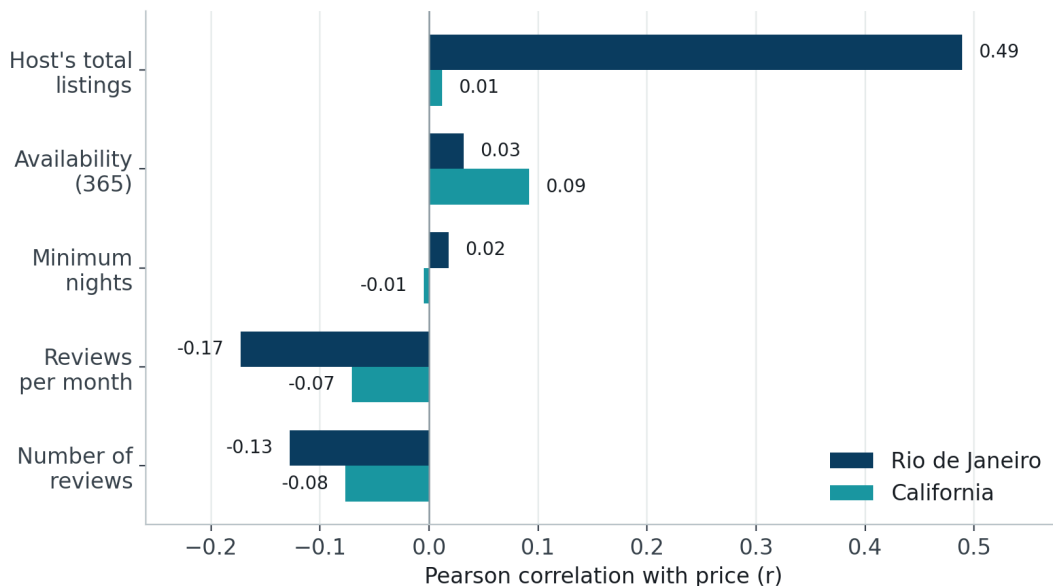


Figure 10. Pearson correlation between price and five listing characteristics, Rio de Janeiro versus Los Angeles County.

Table 11. Correlation Between Price and Each Listing Characteristic

Variable	Rio de Janeiro (Pearson r)	Los Angeles County (Pearson r)	Note
Number of reviews	-0.13	-0.08	
Reviews per month	-0.17	-0.07	
Minimum nights	+0.02	-0.01	LA County: not significant
Availability (365 days)	+0.03	+0.09	
Host's total listings	+0.49	+0.01	Rio Spearman $\rho \approx -0.10$; see Section 7

Exploratory / supporting evidence. All coefficients are statistically significant at $p < 0.01$ except the Los Angeles County minimum-nights correlation. Correlations are bivariate and control for nothing.

Two results deserve a closer look. First, review activity is negatively correlated with price in both markets, consistent with the review-status pattern in Section 6.1, and noticeably stronger in Rio (−0.17) than in Los Angeles County (−0.07). Second, and more strikingly, a host’s total number of listings correlates with price at $r = +0.49$ in Rio but is essentially uncorrelated with price in Los Angeles County ($r = +0.01$). Section 7 shows that the Rio figure is not what it appears to be.

Why correlation alone can mislead

Taken at face value, the Rio host-listings correlation would suggest that hosts with larger portfolios charge substantially more. But this raw correlation controls for nothing. Larger hosts may simply operate more entire-home listings, which Section 4 has already shown command a large premium over private and shared rooms, in which case the independent association between host scale and price, with room type held constant, should be far smaller than $r = 0.49$ suggests.

This is precisely why the analytical workflow moves from single-variable correlation to multivariate regression: listing characteristics are not independent of one another, and a relationship that looks strong in isolation can shrink, vanish, or reverse once other factors are properly accounted for. Section 7 tests this directly, and the answer is the most instructive methodological result in the report.

In the original regression (Table 6), reviews per month is negatively associated with price in both markets once other characteristics are held fixed, but the size of that association differs sharply: a one-unit increase in the reviews-per-month rate (not one additional review, a distinction worth stating plainly since the two are easy to conflate) is associated with a 36.6% lower price in Rio, versus only a 5.7% lower price in Los Angeles County.

Unlike the room-type similarity in Section 4, this cross-market difference does not rest on eyeballing two separate coefficients. It is directly confirmed by a significant reviews-per-month $\times$ market interaction term in the pooled model ($p < 0.001$), reported in full in Section 8. This is a genuinely tested cross-market difference, not a suggestive one.

6.3 A Supplementary Nonlinear Specification

Table 12. Nonlinear Reviews Specification (Supporting)

Market	R ² (linear)	R ² (log(1+reviews))	ΔR^2	ΔAIC
Rio de Janeiro	0.2825	0.3105	+2.8 pp	$\approx -1,412$
Los Angeles County	0.3456	0.3496	+0.4 pp	≈ -200

Exploratory / supporting specification. The large negative ΔAIC indicates a decisive fit improvement in both markets, though the practical gain in Los Angeles County is small.

A supporting, exploratory specification replaces the linear reviews-per-month term with its logged form, $\log(1 + \text{reviews per month})$, which better accommodates the variable’s heavily right-skewed distribution. This nonlinear form improves fit noticeably in Rio (R^2 rises from 0.2825 to 0.3105, a meaningful gain confirmed by a large drop in the model’s AIC) and only marginally in Los Angeles

County (R^2 rises from 0.3456 to 0.3496). The improved fit is consistent with a diminishing, concave relationship: most of the price difference associated with review activity shows up in the initial move from zero reviews to a modest review rate, with the association weakening at higher review-activity levels, which is also what the large zero-review gap in Table 10 would lead one to expect. This report does not claim that Rio's relationship is formally "more nonlinear" than Los Angeles County's, since no direct test of that specific claim was run; the safer, fully tested claim is the linear interaction already reported.

It is also worth flagging precisely how this logged coefficient should and should not be read: a one-unit change in $\log(1 + \text{reviews per month})$ corresponds to roughly tripling the underlying reviews-per-month rate, not to "one additional review", so this specification describes a shape, not a per-review rate, and should never be quoted as "each extra review is associated with an X% price change."

What this does, and does not, show

Every result in this section is an association, not a causal effect, and the data cannot tell us which direction the relationship runs. It is plausible that an unusually high price discourages bookings and therefore limits the reviews a listing can accumulate; it is equally plausible that a newly listed property has simply not had time to collect reviews yet, independent of its price; and a boutique or luxury property with naturally low turnover would look identical to both. Either way, the pattern is a signal worth investigating, not proof that a listing is mispriced.

What this shows and why it matters managerially. A high-priced, unreviewed listing is not evidence of strong demand, and it is not evidence of overpricing either; this dataset cannot distinguish between the explanations above. What it does support is a market-specific screening use: in Rio, the review-activity/price relationship is a much stronger signal, both in the binary comparison and in the regression, and is worth weighting more heavily when flagging listings for closer benchmarking review; in Los Angeles County, the same signal is present but far weaker and should be weighted correspondingly lightly. A platform manager building an automated pricing-anomaly flag, or a host deciding how much weight to put on their own review count when setting expectations, should calibrate that weighting per market rather than applying a single global rule. The practical form of the finding is a benchmarking flag, not a verdict: a listing priced well above its market average with zero recorded reviews is a candidate for closer comparison against similar, already-booked listings before its price is finalised, particularly in Rio.

7. Host Portfolio Scale

Does operating multiple listings go along with a different pricing pattern than operating a single one? This section answers that in three steps of increasing rigour: a raw binary split, a four-tier breakdown, and a room-type-adjusted version of the same tiers, because the answer changes materially at each step. Host-listings count is used throughout strictly as an operational scale indicator, not as a marker of professionalism or service quality, which this dataset has no way to measure.

7.1 The Raw Binary Split

Table 13. Individual versus Multi-Listing Hosts

Measure	Rio: individual (1 listing)	Rio: multi-listing (2+)	LA County: individual	LA County: multi-listing
Listings (n)	20,412	14,964	15,133	17,603
Avg. price (USD)	\$124	\$115	\$175	\$178
Avg. availability (days/yr)	150	202	129	198
Avg. reviews / month	0.28	0.37	1.09	1.26
Price gap vs. individual	reference	-7.3% (t = -4.09, p < 0.001)	reference	+1.7% (t = 1.16, p ≈ 0.25)

A raw, unadjusted split. See Tables 14 and 15 for the finer, room-type-adjusted picture.

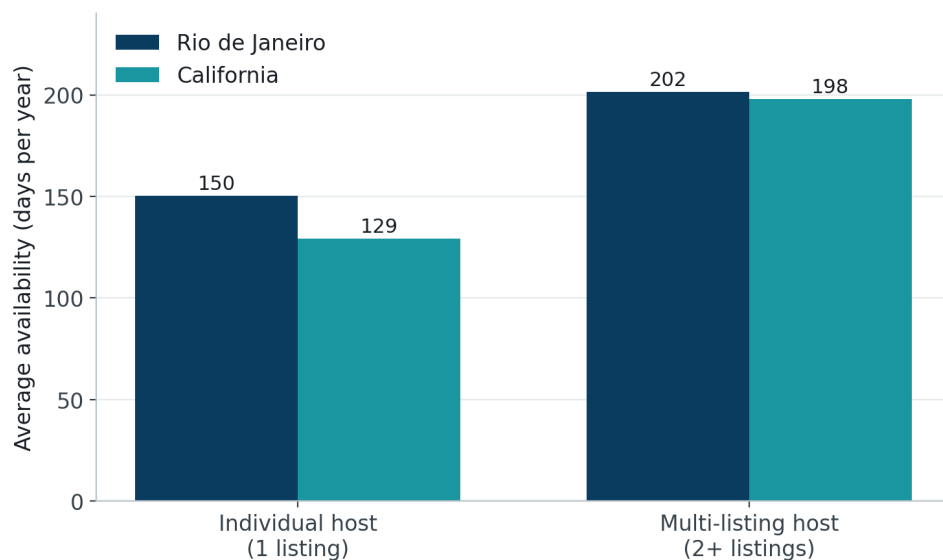


Figure 11. Average availability (days per year) by host type, Rio de Janeiro versus Los Angeles County.

The raw picture already differs by market on price. In Rio, individual-listing hosts price about 7.3% higher on average than multi-listing hosts, a statistically significant gap (t ≈ -4.09, p < 0.001). In Los Angeles County, the same binary comparison shows no significant price gap at all (+1.7%, t ≈ 1.16, p ≈ 0.25); we cannot conclude that multi-listing hosts in Los Angeles County price any differently from individual hosts.

Availability tells a far more consistent story than price does. In both markets, multi-listing hosts keep their properties bookable for substantially more days per year than individual hosts: 202 versus 150 days in Rio (t = 31.4, p < 0.001) and 198 versus 129 days in Los Angeles County (t = 45.0, p < 0.001). This is the one host-scale pattern that clearly generalises across both markets; larger operators run their listings with higher year-round calendar availability, regardless of how that does or does not translate into price. Availability here is a calendar-activity measure, not an occupancy measure: it records how many days a listing was bookable, not how many nights were booked.

7.2 Four Tiers, Raw and Room-Type-Adjusted

Table 14. Host-Scale Raw Four-Tier Means

Market	Tier (listings)	N	Raw mean price (USD)
Rio de Janeiro	1	20,412	\$124.36
Rio de Janeiro	2–4	9,193	\$83.80
Rio de Janeiro	5–9	2,341	\$84.77
Rio de Janeiro	10+	3,430	\$219.77
Los Angeles County	1	15,133	\$175.40
Los Angeles County	2–4	8,992	\$173.07
Los Angeles County	5–9	3,512	\$177.57
Los Angeles County	10+	5,099	\$188.31

Raw means only; always read alongside Table 15, the room-type-adjusted companion.

Table 15. Host-Scale, Room-Type-Adjusted Effects

Market	Tier (vs. single listing)	Adjusted % effect	p-value
Rio de Janeiro	2–4	–15.2%	< 0.001
Rio de Janeiro	5–9	–17.6%	< 0.001
Rio de Janeiro	10+	+10.6%	< 0.001
Los Angeles County	2–4	+0.2% (not significant)	≈ 0.79
Los Angeles County	5–9	–5.3%	< 0.001
Los Angeles County	10+	–8.7%	< 0.001

Adjusted for room-type composition, the mandatory companion to Table 14. Raw and adjusted patterns diverge materially, especially in Los Angeles County.

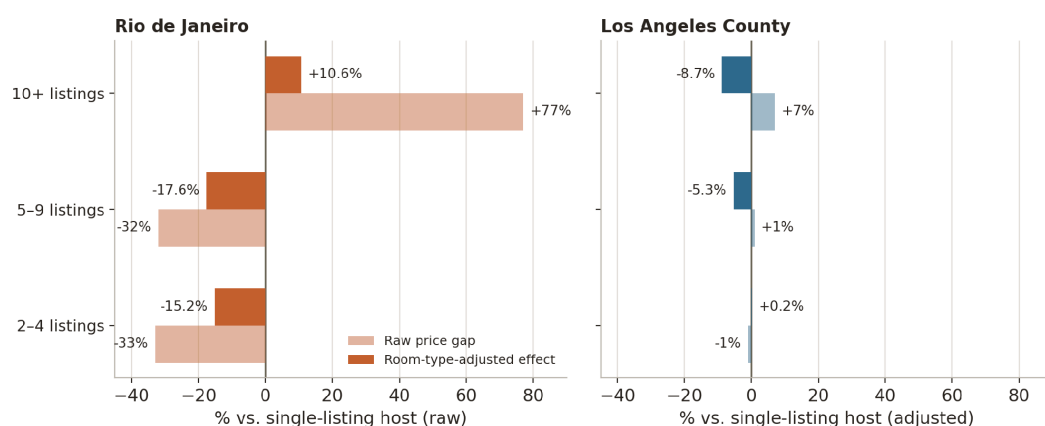


Figure 12. Host-scale price pattern: raw vs. room-type-adjusted. Raw tier averages overstate the effect in Rio and mischaracterize it in Los Angeles County; the adjusted figures are the trustworthy read.

A finer four-tier breakdown (1 listing; 2–4; 5–9; 10+) sharpens the picture, and changes it once room type is accounted for. In raw terms, Rio's 10+-listing tier averages \$219.77 per night, dramatically above the single-listing tier's \$124.36, while the 2–4 and 5–9 tiers actually average lower than single-listing hosts (\$83.80 and \$84.77). But room type is not evenly distributed across these tiers: Rio's 10+-listing hosts are disproportionately entire-home operators (85.6% of their listings, versus 76.8% for single-listing hosts), so part of that raw gap is a room-type composition effect, not a pure scale effect.

Once room type is held fixed, the pattern survives but shrinks substantially: 2–4 listings is associated with a 15.2% lower price than single-listing hosts, 5–9 listings with a 17.6% lower price, and 10+ listings with a 10.6% higher price (all significant at $p < 0.001$). This is a real, non-monotonic pattern; price dips through the middle of the portfolio-size range and rises again only for the largest operators, but its magnitude is considerably smaller than the raw numbers alone would suggest, and the 10+ tier's own raw mean is itself pulled upward by a right-skewed subset of very high-priced listings within that tier (the tier's median price sits far below its mean), so "\$219.77" should not be read as representative of a typical large-portfolio Rio host.

Los Angeles County tells a different story once the same adjustment is applied. The raw tier means already move very little across portfolio size (\$175.40 at one listing, rising only to \$188.31 at 10+), and once room type is held fixed, that small raw rise disappears entirely: 2–4 listings shows no statistically distinguishable difference from single-listing hosts (+0.2%, $p \approx 0.79$), while 5–9 and 10+ listings are both associated with modestly lower prices (–5.3% and –8.7%, both $p < 0.001$). The mild upward drift visible in Los Angeles County's raw averages is attributable to larger-portfolio hosts' room-type mix (10+-listing hosts are 69.5% entire-home versus 66.5% for single-listing hosts), not to any genuine, room-type-independent host-scale pricing effect.

7.3 Reconciling the Correlation, the Continuous Coefficient, and the Tiers

Three different specifications in this report speak to host scale, and they must be read as answering three different questions rather than as conflicting results.

- **The raw Pearson correlation** ($r = +0.49$ in Rio, Table 11) is bivariate and controls for nothing. It is also fragile: the rank-based Spearman correlation for the same pair is slightly negative ($\rho \approx -0.10$), meaning the positive linear correlation is driven by a comparatively small number of extreme high-price, high-listing-count observations, while the typical relationship across the bulk of the distribution runs the other way. Reporting the Pearson figure alone would be actively misleading.
- **The continuous regression coefficient** (+0.63% per additional listing in Rio, –0.15% in Los Angeles County, Table 6) holds room type and the other characteristics fixed and imposes a straight-line relationship across the whole range of portfolio sizes. Both are statistically significant given the sample sizes and both are economically negligible. The collapse from $r = 0.49$ to +0.63% is the clearest demonstration in this report of why the regression stage matters: it separates a real, direct relationship (room type) from one that looked strong only because of a confound.
- **The room-type-adjusted tier analysis** (Table 15) holds room type fixed but allows the relationship to bend. It is the most appropriate specification for this variable, and it reveals

what the linear coefficient averages away: the Rio pattern is non-monotonic, with a mid-range discount and a premium confined to the very largest operators. The continuous coefficient is retained in the main regression as a control; the tier analysis is the basis for every host-scale claim in this report.

Consistent pattern, market-specific price effect

Host scale behaves like a market-specific relationship rather than a universal one. Multi-listing hosts consistently keep listings available for longer in both markets, but whether that goes along with a price discount, a premium confined to the largest operators, or no meaningful difference at all depends on the local market. A pricing rule based on host scale in one market should not be assumed to hold in the other.

What this shows and why it matters managerially. Portfolio-scale strategy should not be copied across these two markets. In Rio, there is a real, if modest, adjusted premium for the very largest operators (10+ listings) and a real discount for mid-scale portfolios, a market-specific pattern worth building into regional guidance for Rio hosts considering scaling up. In Los Angeles County, once room-type composition is accounted for, there is no comparable scale advantage anywhere in the range examined, and a mild scale-related discount at the largest sizes; "go bigger" is not a pricing lever a Los Angeles County host or the platform should promote as such. What does generalise is operational: higher availability among multi-listing hosts in both markets, which is a useful signal of a more actively managed listing and belongs in a benchmarking tool as an operational descriptor rather than a price driver. This entire section should also be read with the endogeneity caveat that runs throughout this report: host-listings count, like reviews and availability, is plausibly set jointly with pricing strategy by the same host decision process, so these patterns describe an association between scale and price, not a demonstrated effect of scale on price.

8. Cross-Market Synthesis

The preceding sections have repeatedly flagged apparent similarities and differences between Rio and Los Angeles County without formally testing most of them. This section closes that gap. A single pooled regression, combining both markets ($N = 68,112$) and including market $\times$ room-type and market $\times$ reviews-per-month interaction terms alongside the same controls used throughout, makes it possible to ask directly: which of the patterns that looked similar or different across the last five sections actually are, in a formal statistical sense? Each interaction term tests a stated hypothesis of the form $H_0: \beta(\text{characteristic, Rio}) = \beta(\text{characteristic, Los Angeles County})$, against H_1 : the two differ. Rejecting H_0 means the relationship is market-specific; failing to reject it means the report has found no evidence that the relationship needs separate calibration by market.

Table 16. Pooled Interaction Model

Term	Formally different across markets?	p-value (HC1)
Market $\times$ Entire home	No	≈ 0.353
Market $\times$ Hotel	Yes	≈ 0.013

Term	Formally different across markets?	p-value (HC1)
Market × Shared room	Yes	< 0.001
Joint room-type × market test (3 restrictions, HC1-robust)	Yes	$F \approx 38.7$, $p < 0.001$
Market × Reviews per month	Yes	< 0.001

Pooled model, $N = 68,112$, HC1 heteroskedasticity-robust standard errors throughout. $F \approx 38.7$ is the HC1-consistent joint statistic used here, not the classical-SE value of 44.4, which uses a different, inconsistent covariance estimator.

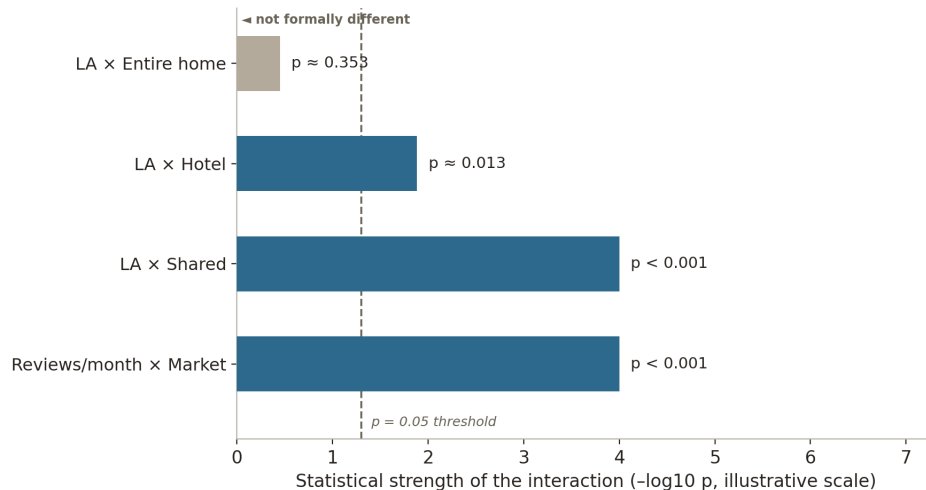


Figure 13. Pooled model: which cross-market differences are formally real? Only the entire-home interaction falls short of the significance threshold; every other tested interaction is formally significant.

Three results answer that question precisely. The market × entire-home interaction term is not statistically significant ($p \approx 0.353$); there is no formal evidence that the entire-home premium differs between Rio and Los Angeles County, which converts Section 4's suggestive similarity into a tested one. The market × hotel interaction term is significant ($p \approx 0.013$), and the market × shared-room interaction term is significant at a much stronger level ($p < 0.001$); both room-type categories genuinely do behave differently across the two markets, a result that would be easy to miss if only the two separate per-market coefficients were compared side by side. Jointly testing all three room-type interaction terms together, using heteroskedasticity-robust standard errors, gives $F \approx 38.7$ ($p < 0.001$), a joint, decisive rejection of the idea that room-type effects are identical across markets overall, even though one specific category (entire-home) is not distinguishable on its own. Separately, the market × reviews-per-month interaction term is also significant ($p < 0.001$), formally confirming the review-activity asymmetry described in Section 6.

It is worth being precise about what the non-significant entire-home interaction does and does not establish. Failing to reject a difference is not the same as proving the two premiums are identical; with samples of this size the test has ample power to detect a difference of managerial consequence, which is what makes the null result informative, but it remains an absence of detected difference rather than a demonstrated equivalence.

Table 17. Summary of Findings by Cross-Market Consistency

Characteristic	Pattern observed	Transfers across markets?	Basis
Room type (entire home)	Entire-home listings price roughly 146% above private rooms in both markets, before and after regression controls and after a neighbourhood control.	Yes: strongest, most transferable relationship	Formally tested (interaction $p \approx 0.353$)
Room type (hotel, shared)	Hotel +31.3% / +68.1%; shared -33.4% / -51.5% relative to private rooms.	No: sizes differ by market	Formally tested ($p \approx 0.013$; $p < 0.001$)
Review activity	Negative association with price in both markets; roughly six times steeper in Rio on the regression coefficient and about three times larger on the review-status gap.	No: much stronger in Rio	Formally tested (interaction $p < 0.001$)
Currency-adjusted price level	Los Angeles County is about 47% more expensive than Rio once both are expressed in USD.	Not applicable: a level comparison, not a relationship	Descriptive
Neighbourhood	Top neighbourhoods price 3–4× their market average in both cities; controlling for neighbourhood shrinks the entire-home premium only modestly.	Directionally yes; the cross-market difference was not formally tested	Descriptive plus regression control
Host scale → price	Adjusted premium only for Rio's largest operators (+10.6%); flat to mildly negative at every tier in Los Angeles County.	No: market-specific	Per-market adjusted estimates; not jointly tested
Host scale → availability	Multi-listing hosts show materially higher availability in both markets (202 vs. 150 days; 198 vs. 129 days).	Yes	Per-market t-tests

The "Basis" column is deliberate: only the first three rows rest on a formal cross-market test. The remaining rows are descriptive or based on separately estimated per-market results and carry a correspondingly weaker standard of evidence.

Put together, these results draw a precise line between what transfers across these two markets and what does not. Entire-home pricing structure transfers; secondary room-type categories and review-based pricing dynamics do not. This is a more nuanced and more useful conclusion than either "the two markets are the same" or "the two markets are different"; it identifies exactly which pieces of the pricing story are shared and which are not, and it does so using the same formal-testing discipline throughout: a significance test on the difference itself, not a comparison of two separate p-values.

What this shows and why it matters managerially. This is the single most decision-relevant section of the report for a platform or regional manager, because it converts every "looks similar" or "looks different" impression from earlier sections into a tested yes-or-no answer. A manager can build one shared entire-home pricing or benchmarking standard that applies credibly to both markets, but must build separate, market-specific rules for hotel and shared-room pricing, and for how heavily review activity is weighted as a benchmarking signal. No interaction beyond these tested terms is claimed here; host-scale differences across markets, for instance, are described in Section 7 using separate per-market adjusted estimates, not a formal joint cross-market test, and should be read with that more limited standard of evidence in mind.

9. Managerial Implications

Two decision-makers read this report differently. A platform or regional manager cares about which pricing patterns can be standardized across markets and which need local rules, because every standardized rule saves operational effort but every wrongly-standardized rule risks a bad recommendation in the market where it does not apply. An individual host or listing manager cares about how to benchmark and position one property, because they cannot experiment with price the way a platform can experiment with policy. Every recommendation below is traceable to a specific, already-reported result; none of it is generic pricing advice.

9.1 Room Type: The Primary Segmentation Axis

Data result: Entire-home listings carry a large premium over private rooms in both markets (Rio +147.1%, Los Angeles County +145.7%), and a formal interaction test shows this premium does not significantly differ between markets ($p \approx 0.353$).

Analytical interpretation: Room type is the single most stable, most cross-market-general structural characteristic identified anywhere in this analysis.

Platform manager insight: Room-type category is the one segmentation variable that can be used as a shared, cross-market benchmarking dimension without market-specific adjustment.

Platform manager action: Build the platform's cross-market listing-comparison and pricing-guidance tools around room type as the primary axis, with market-specific refinements layered on afterward rather than built in from the start. Note that the same tool will be applied to differently-weighted inventories: 71.6% of Rio's listings are entire homes against 63.2% in Los Angeles County.

Platform manager trade-off: A shared room-type standard says nothing about the right absolute price in each market; Los Angeles County's baseline price level is substantially higher than Rio's throughout, so "same relative premium" does not mean "same price."

Host insight: A listing's basic category does a large share of the pricing work before location, reviews, or anything else is considered.

Host action: When comparing a listing to others for pricing or positioning purposes, first restrict the comparison set to the same room type, in the same market, before refining further. Comparing a private room against a citywide average dominated by entire homes will systematically misstate a competitive price.

Host trade-off: This is an observed association from cross-sectional data, not a demonstrated effect of switching categories; room type is very likely correlated with unmeasured attributes (size, amenities, build quality) that this dataset cannot separate out.

9.2 Neighbourhood: The Local Benchmark

Data result: Top-priced neighbourhoods price at roughly 3.2× the city average in Rio and roughly 4.0× the county average in Los Angeles County (descriptive), while the entire-home premium shrinks only modestly once the top 15 neighbourhoods per market are statistically controlled for (Rio 147.1% → 136.0%; Los Angeles County 145.7% → 140.2%).

Analytical interpretation: Location adds genuine explanatory information and produces very wide within-market dispersion, but it does not substitute for or explain away the room-type story.

Platform manager insight: A location-only benchmarking rule would systematically under- or over-value listings, because it would be missing the larger room-type effect running alongside it, and a market-average benchmark would misprice both ends of the neighbourhood distribution badly.

Platform manager action: Any neighbourhood-level pricing guidance or "comparable listings" tool should condition on room type and neighbourhood jointly, not neighbourhood alone, and should use granular neighbourhood rather than the broad zone field, whose Los Angeles County effect is significant but small (about 15% between the City of Los Angeles and unincorporated areas) and which is unpopulated for Rio entirely.

Platform manager trade-off: The neighbourhood model used here is a statistical control on a coarse geography (top-15 named areas plus a large pooled "other" category), not a fine-grained or causal account of what specifically makes a location valuable.

Host insight: "My neighbourhood explains my price" is, at best, a partial explanation; room type still matters substantially after location is accounted for.

Host action: Benchmark against comparable listings that match on both room type and neighbourhood tier, not neighbourhood alone, and never against a citywide average if the property sits at either extreme of the local price distribution.

Host trade-off: The neighbourhood premiums in Table 7 are descriptive: an expensive area may partly be an entire-home-heavy area, so a neighbourhood average is a benchmark, not a proven independent cause of a higher price.

9.3 Review Activity: A Market-Calibrated Signal

Data result: Reviews per month is negatively associated with price in both markets, far more strongly in Rio (-36.6% per unit) than in Los Angeles County (-5.7%), and this cross-market difference is formally confirmed by a significant interaction term ($p < 0.001$). The binary review-status gap points the same way (98.8 index points in Rio versus 32.0 in Los Angeles County), and 41.6% of Rio listings have no reviews at all, against 20.9% in Los Angeles County.

Analytical interpretation: The review-activity/price relationship is a genuinely market-specific dynamic, not a universal one, and one of the clearest examples in this report of a difference that formal testing actually confirms rather than merely suggests.

Platform manager insight: Review-activity-based pricing-anomaly flags (for example, "high price, very low review activity") carry much stronger signal value in Rio than in Los Angeles County.

Platform manager action: If review activity is used as an input to any automated flagging, ranking, or anomaly-detection system, weight it more heavily for Rio listings and correspondingly more lightly for Los Angeles County listings, rather than applying one global weight.

Platform manager trade-off: Reviews per month is an engagement/activity proxy, not a bookings or demand measure: a flagging system built on it will surface newly listed and niche properties alongside any genuinely mispriced ones, and cannot distinguish between those explanations on its

own. In Rio this matters at scale, since over 40% of listings would enter such a flag's unreviewed pool.

Host insight: A listing with a high price and few or no reviews is not, by itself, proof of anything: strong demand, weak demand, and simple newness can all look identical on this one variable.

Host action: A Rio host with low review activity and an above-market price has more reason to treat that combination as a flag worth investigating than a Los Angeles County host with the same pattern, given how much more strongly the two are associated in Rio.

Host trade-off: Reviews accumulate over time and depend on guest willingness to leave feedback, not on host effort alone; a new or recently relisted property will show the same pattern as a genuinely overpriced one, for reasons that have nothing to do with pricing strategy.

9.4 Host Portfolio Scale: Secondary and Operational

Data result: Once room type is held fixed, Rio shows a real, non-monotonic pattern (2–4 listings $\approx -15.2\%$, 5–9 $\approx -17.6\%$, 10+ $\approx +10.6\%$, all $p < 0.001$); Los Angeles County shows essentially no adjusted host-scale effect (2–4 not significant, 5–9 $\approx -5.3\%$, 10+ $\approx -8.7\%$). Multi-listing hosts run materially higher availability in both markets.

Analytical interpretation: Only the largest Rio operators show an adjusted price association above single-listing hosts; every other tier in both markets shows either no meaningful adjusted difference or a discount. The one genuinely cross-market host-scale pattern is operational, not financial.

Platform manager insight: "Operating more listings is associated with charging more" is not a transferable rule; it holds, in a limited and adjusted sense, only for the largest Rio operators, and does not hold in Los Angeles County at any scale examined here.

Platform manager action: Treat host-scale segmentation as market-specific pricing guidance, and use scale primarily as an operational descriptor (availability, active management) in benchmarking tools rather than as a price driver. Do not extend a Rio-derived "scale premium" narrative into Los Angeles County host communications or platform policy.

Platform manager trade-off: Host-listings count is an operational scale indicator only, not a validated measure of professionalism, service quality, or business sophistication, and it is plausibly set jointly with pricing strategy by the same host decisions rather than independently of them.

Host insight: Expanding a portfolio is not, on this evidence, a reliable route to higher per-listing prices in either market except at the very largest scale in Rio specifically, and even that pattern is considerably smaller once room-type composition is accounted for than a first glance at raw averages would suggest.

Host action: A host weighing whether to scale up should not treat "more listings" as a pricing strategy in itself; if scaling, the room-type mix of the added listings matters more than the count alone.

Host trade-off: This is a cross-sectional association among currently active hosts of different sizes, not a before/after comparison of what happens to price when a specific host scales up; the two are not the same claim.

9.5 Cross-Market Transfer: What Can and Cannot Be Standardized

Data result: The pooled interaction model shows the entire-home premium is not formally different across markets ($p \approx 0.353$), while the hotel premium ($p \approx 0.013$), the shared-room premium ($p < 0.001$), and the review-activity relationship ($p < 0.001$) all are.

Analytical interpretation: The two markets share one genuinely common structural feature, the size of the entire-home advantage, while diverging on secondary room-type categories and on how strongly engagement relates to price.

Platform manager insight: Cross-market standardization is not an all-or-nothing choice; this analysis identifies exactly which pieces of pricing logic can be shared and which cannot.

Platform manager action: Build one shared entire-home benchmarking standard usable across markets, alongside separate, market-calibrated rules for hotel/shared-room pricing and for review-activity-based signals. Re-test any adjustment developed in one market before applying it to another.

Platform manager trade-off: Only the specific terms tested here carry this formal warrant, host-scale patterns, for instance, were not jointly tested across markets and should not be assumed to carry the same "formally different" status without a comparable test.

Host insight: A host relocating a strategy from one of these markets to the other can trust the entire-home positioning logic to travel; they should not assume the same about hotel/shared-room pricing norms or about how much weight to put on review activity.

Host action: Treat "what similar hosts do in the other market" as a reasonable starting point only for entire-home listings; treat it with real caution for other room types or for review-based decisions.

Host trade-off: Even where a pattern is confirmed as formally similar, that says nothing about whether the underlying reasons for the pattern are the same in both markets, only that its statistical size is indistinguishable.

9.6 Cross-Market Price Levels: Convert Before Concluding

Data result: Rio averages roughly R\$620 per night and Los Angeles County roughly \$177; converted at 5.15 BRL/USD, Rio averages about \$120 and Los Angeles County is about 47% more expensive. The median gap is wider still (1.84×).

Analytical interpretation: Currency conversion does not merely rescale the comparison; without it, the conclusion about which market is the premium one is reversed.

Platform manager insight: Any comparison of pricing, revenue targets, or host performance across countries is invalid before currency harmonisation, and any benchmark built on means rather than medians will be distorted by both markets' heavy right tails.

Platform manager action: Standardise all cross-market reporting on a common currency and on median-based benchmarks, and state the conversion rate and date alongside any published figure.

Platform manager trade-off: A single exchange rate is a snapshot; it fixes the comparison at one moment and does not account for purchasing-power differences between the two economies, which this dataset cannot address.

Host insight: Comparing one's own nightly rate to figures quoted from another country's market is meaningless without conversion, and misleading even after it if the comparison uses market averages rather than same-segment medians.

Host action: Benchmark within the local market and the local currency first; use cross-market comparisons only for structural questions, such as relative room-type positioning, not for absolute price levels.

Host trade-off: Even a correct currency conversion compares price levels, not value; it says nothing about the cost base, regulation, or guest expectations a host faces in either market.

9.7 Recommendation Summary

Table 18 consolidates the preceding six subsections into a single decision matrix. Priority reflects two things together: how large the underlying effect is, and how strong the evidence behind it is. A recommendation resting on a formally tested result and a large effect outranks one resting on a descriptive pattern or a small effect, however intuitive the latter may be.

Table 18. Recommendation Summary Matrix

Priority	Recommendation	Primary owner	Supporting evidence	Evidence strength
1	Segment on room type first in every pricing, benchmarking, and comparison tool, in both markets.	Platform and host	Tables 4, 5, 6; interaction $p \approx 0.353$	Strong: large effect, formally tested
2	Convert to a common currency and benchmark on medians before any cross-market comparison.	Platform	Table 3; Figure 1	Strong: direction of conclusion reverses without it
3	Layer neighbourhood on top of room type; never benchmark against a citywide average alone.	Platform and host	Tables 7, 9; Figures 7, 8	Moderate to strong: descriptive ranking plus regression control
4	Calibrate review-based pricing-anomaly flags separately per market, weighting them more heavily in Rio.	Platform	Tables 6, 10; interaction $p < 0.001$	Strong: formally tested cross-market difference
5	Treat host scale as an operational descriptor rather than a pricing lever; do not export a Rio scale narrative to Los Angeles County.	Platform	Tables 13, 14, 15	Moderate: per-market adjusted estimates, not jointly tested
6	Ignore minimum nights and availability as pricing levers in both markets.	Host	Table 6	Moderate: significant in Los Angeles County but economically negligible

Priority is a joint judgement on effect size and evidence strength, not on ease of implementation. Every row is traceable to a numbered table in the main body; no recommendation rests on evidence not reported in this document.

10. Limitations and Future Research

10.1 Limitations

This analysis is genuinely informative about the associations present in this specific dataset, but its boundaries deserve the same explicit, evidence-tied treatment as its findings.

Observational, cross-sectional design. Every regression and ANOVA result in this report describes a conditional association observed in a single snapshot of listings, not a causal effect. There is no experiment, no random assignment of listing characteristics, and no repeated observation of the same listings over time. This limitation applies to every finding in Sections 3 through 9 without exception, and is the reason this report consistently uses language such as "associated with" rather than "causes," "drives," or "determines" throughout.

Omitted property attributes. The assignment's variable set does not include listing size, bedroom count, amenities, or guest rating scores, all plausible price drivers this analysis simply cannot observe. This is very likely a major contributor to the unexplained variance in every regression reported here: roughly 72% of the variance in log-price in Rio and roughly 65% in Los Angeles County (for the original models; slightly less once neighbourhood is added) is not accounted for by room type, reviews, minimum nights, availability, and host-listings count combined. This specifically threatens any temptation to read the reported R^2 values as evidence that this analysis has identified "the" determinants of price; it has identified some of the strongest observable associations, not a complete account.

No bookings or occupancy data. Reviews per month and availability over 365 days are used throughout this report strictly as activity and calendar proxies. Neither is a direct measure of demand, bookings, or occupancy, and this dataset provides no way to construct one. This limitation most directly threatens Section 6's review-activity findings and Section 7's host-scale findings, both of which lean on these variables, and is why both sections are written in terms of engagement and calendar activity, not success or demand.

Neighbourhood modelling constraints. The geographic control used in Section 5.3 covers only the fifteen largest named neighbourhoods per market, with everything else pooled into a single "other" category. The descriptive ranking in Section 5.1 covers a wider set but controls for nothing. This threatens the precision, though not the qualitative direction, of Section 5's location-control claims: a finer geography might shrink the entire-home premium further, or might not; this analysis cannot say. The broader zone field offers no help for Rio, where it is unpopulated for every listing.

Geographic coordinates are not used for spatial modelling. Both datasets include latitude and longitude for every listing. This report uses them only implicitly, through the named neighbourhood a listing falls into; it does not derive continuous distance-based features such as distance to the coastline or to the city centre. Such measures could reveal geographic gradients, plausibly important in a coastal leisure market like Rio, that named neighbourhood boundaries do not fully capture.

Endogeneity of reviews, availability, and host-listings count. All three of these variables are plausibly set jointly with pricing strategy by the same host decision process, rather than being independent, exogenous inputs. A host who prices aggressively may also manage availability and review accumulation differently than a host who does not; a large-portfolio host's price and their

listing count may both reflect an underlying business model this dataset cannot observe directly. This threatens any temptation to read Sections 6, 7, or 9's findings as identifying levers a host or platform can pull in isolation; they describe associations among variables that likely move together, not independent causes.

Outlier-treatment sensitivity. The entire-home price effect central to Sections 4, 5, and 8 moves by a real, non-trivial amount depending on how aggressively extreme prices are trimmed before analysis.

Table 19. Outlier-Sensitivity Robustness (Supporting)

Market	Entire-home % at 99th pct. trim	At 99.5th pct. trim	No trim
Rio de Janeiro	+147.1%	+148.7%	+152.3%
Los Angeles County	+145.7%	+151.3%	+156.6%

Direction and broad magnitude are robust across trimming rules; the exact percentage is not, most notably for Los Angeles County, where it moves by nearly 11 percentage points.

The direction and rough magnitude of the effect are robust (it never drops below +145% or reverses sign), but the specific percentage quoted throughout this report should be understood as sensitive to this modelling choice, particularly for Los Angeles County, and not as a fixed constant of either market.

Single-snapshot dataset. This dataset captures listings at one point in time, with no seasonal or year-over-year dimension. Every finding in this report describes the market as it appeared in this snapshot; none should be read as a claim about how these relationships evolve over time or across seasons.

Exploratory status of supporting analyses. The neighbourhood-adjusted model and the pooled cross-market interaction model were both specified as direct extensions of the two original, pre-specified per-market regressions, and are treated with the same confidence. The correlation matrix, the nonlinear reviews specification, the descriptive neighbourhood ranking, and the outlier-sensitivity check, by contrast, were developed after the core analysis, in response to gaps identified along the way; they are reported here as exploratory, supporting evidence, not as pre-registered confirmatory tests, and the relevant sections flag this status explicitly wherever these specific results are used.

10.2 Future Research

Four extensions would address the most consequential of the limitations above, in roughly descending order of expected value:

- **Add richer listing-quality data:** amenities, bedroom count, review scores, and Superhost status where available, to address the omitted-attribute problem that most plausibly accounts for the two-thirds of log-price variance this model leaves unexplained.
- **Extend the pooled interaction framework** to terms this report tested only per-market, particularly host scale and neighbourhood, so that those cross-market comparisons can be made to the same formal standard as the room-type and review-activity ones in Section 8.

- **Track listings over time** to study seasonality and to test whether price changes are followed by changes in review accumulation, which would help clarify the direction of the review-price relationship that Section 6 can only describe as an association.
- **Derive distance-based features from latitude and longitude:** distance to the coastline, to the city centre, or to major attractions, to test whether continuous geographic measures explain price variation beyond what named neighbourhoods capture.

11. Conclusion

This report set out to answer a specific, two-part question: which observable listing characteristics are most strongly associated with Airbnb price differences in Rio de Janeiro and Los Angeles County, and do these relationships look similar or different across the two markets, with "different" claimed only where a formal test supports it. The answer, worked through Sections 3 to 8, is neither "everything transfers" nor "nothing transfers."

Room type, and specifically the large premium commanded by entire-home listings over private rooms, is the one finding that is both large and genuinely cross-market-general: it holds in both cities, survives controlling for neighbourhood, and, uniquely among the room-type categories tested, shows no formally significant difference in size between the two markets. Everything else in this analysis is more market-specific than that headline result might suggest. Hotel and shared-room pricing behave differently in the two cities. Review activity is a far stronger price signal in Rio than in Los Angeles County. Host portfolio scale shows a real, if modest, adjusted premium only for the very largest Rio operators, and essentially no comparable pattern anywhere in Los Angeles County, though multi-listing hosts run materially higher availability in both.

Two methodological lessons run alongside the substantive ones and are worth as much to a manager. The first is that a cross-market comparison is only as trustworthy as the currency conversion behind it: on raw figures, Rio would appear to be the premium market; correctly converted, Los Angeles County is about 47% more expensive. The second is that a strong bivariate correlation can be almost entirely an artefact of composition; Rio's $r = 0.49$ between host scale and price collapses to a negligible adjusted effect once room type is held constant, because larger hosts simply list more entire homes. Both are reminders that the analytical step, not the headline number, is what makes a finding usable.

The honest boundary around all of this is worth restating plainly. This is an association-based, cross-sectional analysis of two markets at a single point in time, built on the variables this assignment made available; it identifies where price differences show up and how strongly, not why they exist, and it is a benchmarking tool for a manager or host trying to understand where a listing or a market segment sits, not a formula for computing an "optimal" price or a substitute for local judgment. Used that way, as a disciplined starting point for comparison with its limitations taken as seriously as its findings, this analysis offers a genuinely useful, evidence-based answer to the question a platform manager and an individual host each actually face: not "what is the right price," but "what does this market currently reward, and does that reward structure travel to the next market over."

References

Course lecture material, Marketing Analytics (MS 491), Lecture 7: practical session on extracting managerial insights from Airbnb listings data. Indian Institute of Technology Gandhinagar.

Airbnb listings data for Rio de Janeiro, Brazil and Los Angeles County, California, provided for this assignment. Variables used: price, room type, neighbourhood, latitude and longitude, minimum nights, number of reviews, reviews per month, host listings count, and availability over 365 days.

Approximate USD/BRL exchange rate of 5.15, reflecting rates observed in September 2026, used for all currency conversion in this report.

Analysis conducted in Python using the pandas, scipy, and statsmodels libraries.

Appendix A: Full Regression Output, Rio de Janeiro

Dependent variable: $\log(\text{price})$. Reference category for room type: private room. Standard errors in this appendix are classical (heteroskedasticity-unadjusted) OLS standard errors, as produced by the software. The significance statements in the main body rest on HC1 heteroskedasticity-robust standard errors, as described in Section 2.4; the point estimates are identical under both.

Variable	Coefficient	Std. error	t-stat	p-value	95% CI
Intercept	5.2580	0.0108	485.79	< 0.001	[5.2368, 5.2792]
Entire home / apt	0.9047	0.0107	84.33	< 0.001	[0.8837, 0.9258]
Hotel room	0.2726	0.0687	3.97	< 0.001	[0.1380, 0.4073]
Shared room	-0.4072	0.0328	-12.43	< 0.001	[-0.4713, -0.3429]
Reviews per month	-0.4552	0.0075	-60.53	< 0.001	[-0.4700, -0.4405]
Minimum nights	-0.00005	0.00021	-0.24	0.807	[-0.00046, 0.00036]
Availability (365)	0.00004	0.00003	1.21	0.226	[-0.00002, 0.00010]
Host's total listings	0.0063	0.0002	41.40	< 0.001	[0.0060, 0.0066]

Table A1. Rio de Janeiro coefficient estimates.

Model statistic	Value
Observations	35,376
R ²	0.2825
Adjusted R ²	0.2823
F-statistic	1,988.9 (p < 0.001)

Table A2. Rio de Janeiro model fit.

Appendix B: Full Regression Output, Los Angeles County

Dependent variable: $\log(\text{price})$. Reference category for room type: private room. Standard errors are classical (heteroskedasticity-unadjusted) OLS standard errors, as produced by the software; see the note in Appendix A.

Variable	Coefficient	Std. error	t-stat	p-value	95% CI
Intercept	4.2319	0.0080	528.85	< 0.001	[4.2162, 4.2475]
Entire home / apt	0.8991	0.0080	112.13	< 0.001	[0.8834, 0.9148]
Hotel room	0.5195	0.0553	9.40	< 0.001	[0.4111, 0.6279]
Shared room	-0.7243	0.0206	-35.13	< 0.001	[-0.7647, -0.6839]
Reviews per month	-0.0589	0.0023	-25.66	< 0.001	[-0.0635, -0.0544]
Minimum nights	-0.0016	0.0001	-13.39	< 0.001	[-0.0018, -0.0014]
Availability (365)	0.00058	0.00003	22.35	< 0.001	[0.00053, 0.00063]
Host's total listings	-0.0015	0.0002	-8.69	< 0.001	[-0.0019, -0.0012]

Table B1. Los Angeles County coefficient estimates.

Model statistic	Value
Observations	32,736
R ²	0.3456
Adjusted R ²	0.3454
F-statistic	2,468.9 (p < 0.001)

Table B2. Los Angeles County model fit

Appendix C: One-Way ANOVA Source Tables

Full variance decomposition behind the raw-price F-statistics reported in Table 5 of Section 4.2. The F-statistic is the ratio of the between-groups mean square to the within-groups mean square. Grouping factor: room type (four categories). Note that the sums of squares are not comparable across the two markets, because Rio's figures are computed on its original BRL price scale.

Market	Source of variation	Sum of squares	df	Mean square	F	p-value
Rio de Janeiro	Between groups (room type)	1,707,340,981	3	569,113,660	562.1	< 0.001
Rio de Janeiro	Within groups (residual)	35,816,806,489	35,372	1,012,575		
Los Angeles County	Between groups (room type)	166,675,119	3	55,558,373	1,112.0	< 0.001
Los Angeles County	Within groups (residual)	1,635,421,256	32,732	49,964		

Table C1. One-way ANOVA source table, price by room type, both markets.

The log-price versions of these tests, reported alongside the raw-price statistics in Table 5 (Rio $F \approx 2,359$; Los Angeles County $F \approx 5,180$), are the more appropriate ones given the unequal variance

across room-type categories on the raw price scale. Both versions reject the null hypothesis of equal room-type means decisively in both markets.

Appendix D: Raw Software Printout

Appendices A and B present the same coefficients shown here, reformatted for readability. This appendix reproduces the unedited output exactly as generated by the statistical software (Python, statsmodels), as direct evidence of where the numbers in the main body come from. Room-type variables are coded as 1/0 indicators, with private room as the omitted reference category, mathematically identical to the reference-category notation used in the main text. Covariance type is "nonrobust" in these printouts; the HC1-robust re-estimation described in Section 2.4 leaves the coefficients unchanged and affects only the standard errors and the significance statistics derived from them.

D.1 Rio de Janeiro

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=====
                        OLS Regression Results
=====
Dep. Variable:          log_price      R-squared:                0.282
Model:                  OLS            Adj. R-squared:          0.282
Method:                 Least Squares   F-statistic:             1989.
Date:                   Thu, 17 Sep 2026 Prob (F-statistic):       0.00
Time:                   11:00:38        Log-Likelihood:          -45321.
No. Observations:       35376          AIC:                    9.066e+04
Df Residuals:           35368          BIC:                    9.073e+04
Df Model:                7
Covariance Type:        nonrobust
=====
                        coef      std err          t      P>|t|      [0.025      0.975]
-----
const                    5.2580      0.011     485.793      0.000      5.237      5.279
Entire_home_apt          0.9047      0.011     84.331      0.000      0.884      0.926
Hotel_room               0.2726      0.069      3.969      0.000      0.138      0.407
Shared_room             -0.4071      0.033    -12.427      0.000     -0.471     -0.343
reviews_per_month       -0.4552      0.008    -60.532      0.000     -0.470     -0.440
minimum_nights          -5.061e-05      0.000     -0.244      0.808     -0.000      0.000
availability_365        3.659e-05      3.02e-05      1.210      0.226    -2.27e-05      9.59e-05
calculated_host_listings_count 0.0063      0.000     41.399      0.000      0.006      0.007
=====
Omnibus:                 1434.836   Durbin-Watson:           1.768
Prob(Omnibus):           0.000     Jarque-Bera (JB):        1614.667
Skew:                    0.513     Prob(JB):                0.00
Kurtosis:                3.207     Cond. No.:               3.43e+03
=====

```

Notes:

- [1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
- [2] The condition number is large, 3.43e+03. This might indicate that there are strong multicollinearity or other numerical problems.

D.2 Los Angeles County

```

=====
                        OLS Regression Results
=====
Dep. Variable:          log_price      R-squared:                0.346
Model:                  OLS            Adj. R-squared:          0.345
Method:                 Least Squares   F-statistic:             2469.
Date:                   Thu, 17 Sep 2026 Prob (F-statistic):       0.00
Time:                   11:00:38        Log-Likelihood:          -32986.
No. Observations:       32736          AIC:                    6.599e+04
Df Residuals:           32728          BIC:                    6.605e+04
Df Model:                7
Covariance Type:        nonrobust
=====
                        coef      std err          t      P>|t|      [0.025      0.975]
-----
const                    4.2319      0.008     528.853      0.000      4.216      4.248
Entire_home_apt          0.8991      0.008    112.130      0.000      0.883      0.915
Hotel_room               0.5195      0.055      9.396      0.000      0.411      0.628
Shared_room             -0.7242      0.021    -35.126      0.000     -0.765     -0.684
reviews_per_month       -0.0589      0.002    -25.660      0.000     -0.063     -0.054
minimum_nights          -0.0016      0.000    -13.390      0.000     -0.002     -0.001
availability_365        0.0006      2.61e-05     22.350      0.000      0.001      0.001
calculated_host_listings_count -0.0015      0.000     -8.693      0.000     -0.002     -0.001
=====
Omnibus:                 5374.531   Durbin-Watson:           1.876

```

Prob(Omnibus):	0.000	Jarque-Bera (JB):	10007.929
Skew:	1.035	Prob(JB):	0.00
Kurtosis:	4.748	Cond. No.	3.31e+03

=====

Notes:

- [1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
- [2] The condition number is large, 3.31e+03. This might indicate that there are strong multicollinearity or other numerical problems.